Image Resampling Detection via Spectral Correlation with False Alarm Control

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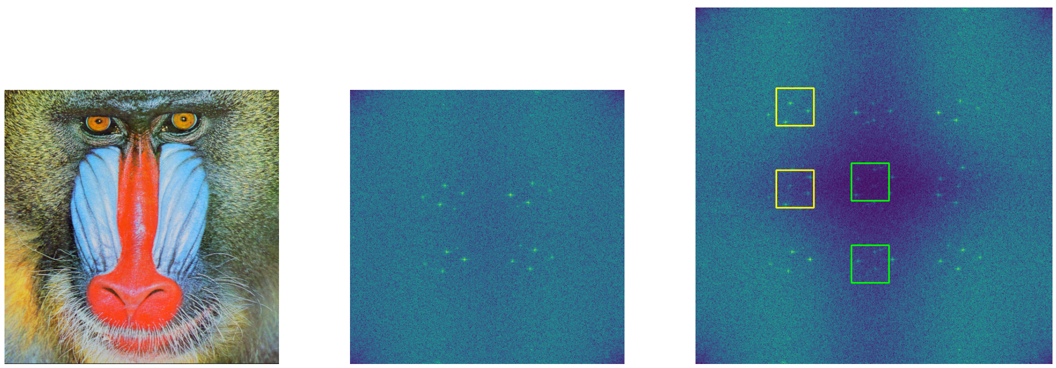


Fig. 1: We detect the anomalous correlations between small patch pairs in the residual spectrum of a resampled image. Left: the original "baboon" image in 512×512. Middle: the spectrum of the residual of the original image by the rank transform filter. Right: the spectrum of the residual of the image upsampled to 665×665 using bicubic interpolation. In the spectrum on the right, two pairs of patches separated by $d = (-512 \bmod 665) = 153$ components along the height are highly correlated. This leaves a hint for resampling detection.

***Abstract*—The detection of image resampling is a critical task in digital forensics, as it is essential for identifying manipulated media and verifying authenticity. Traditional approaches mostly rely on the correlations in the spatial domain caused by resampling, while the correlation in the Fourier domain related to the spectrum folding phenomenon during resampling is overlooked. In this work, we conduct a theoretical study of this phenomenon, and demonstrate that the spectral correlations caused by resampling are exploitable for image resampling detection. We propose an unsupervised method to detect anomalous spectral correlations, which consists of extracting image residuals to suppress natural correlations caused by the image content, and validating remaining significant correlations of resampling using an *a contrario* framework. The proposed method is suitable for images with preserved aspect ratio, even if they are JPEG-compressed. Evaluated in diverse scenarios—including resampled images with different anti-aliasing filters, scaling factors and JPEG compression levels and in-the-wild resampled images from social media-the proposed method demonstrates comprehensive superiority over the existing methods, and shows robustness to different classical and AI-based interpolating filters and image sizes. Code is available at https://github.com/li-yanhao/ird.**

***Index Terms*—image resampling detection, resizing detection, interpolation, image forensics, spectrum folding**

I. Introduction

Image resampling is a common operation in digital image processing. It involves reconstructing the values on a new sampling grid from the pixels of the original grid using interpolation techniques, such as nearest-neighbor, bilinear, bicubic, or Lanczos interpolation. Most images are, for example, rescaled before being sent through social networks. Rescaling is also often involved in image forgery. The detection of resampling traces is therefore crucial to recover the editing history of an image, which is, for example, useful for downstream image authentication tasks [1].

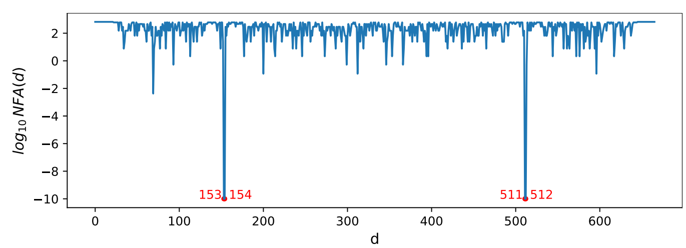


Fig. 2: The number of false alarms (NFAs) for all the possible distances d calculated on the upsampled "baboon" image in Fig. 1. Note that the height of the original image is $M_1 = 512$ and the height of the upsampled image is $N_1 = 655$. According to Sec. III-B, the significant NFAs should appear at distances $d = M_1 (\mathrm{mod}\, N_1) = 512$ and $d = -M_1 (\mathrm{mod}\, N_1) = 153$.

Many forensic algorithms base their analysis on artefacts found early in the imaging pipeline [2–10]; as such they can fail or can even yield false positives when the image no longer has its original size. As shown in Karageogiou et al. [11], AI-generated images are extremely difficult to detect when they have been resized, especially when using spectral artefacts that move with resampling [12, 13]. Knowing whether an image has been resampled is thus necessary to have a correct interpretation of these methods. Used locally, resampling detection could also be used by itself to detect local tampering, for instance if an object has been downscaled prior to splicing.

Beyond image forensics, many image processing steps modify the noise of an image, especially when they involve interpolation steps [14–16]. Resampling an image — and even more so upsampling it — similarly induces correlated noise in the image. Knowing that an image has been resampled, and ideally which resampling factor, would be very useful information to pass to denoising algorithms when trying to restore images found in the wild [17].

Image resampling detection has been a subject of extensive research in the literature over the years. Its first prominent method dates back to Popescu and Farid [1], who applied the expectation-maximization (EM) algorithm to determine the correlation between neighboring pixels, resulting in a correlation probability map (p-map). The method is able to detect resampled images after rotation, and was extended for forgery localization purposes by analyzing the incoherence of resampling traces across the image. Later, Kirchner [18] accelerated the computation of the p-map by replacing the time-consuming EM estimation of image filtering weights by a linear filter with preset coefficients. Furthermore, they presented an automatic approach to detect anomalies in the p-map, and studied the relationship between the geometric transformations of the image and the position of anomalous peaks in the p-map's spectrum. Thereafter, Kirchner *et al.* [19] studied how affine transformations of pre-compressed images affect p-map-based detectors and presented a variant of p-map to detect resampling on pre-compressed images. Nguyen and Katzenbeisser [20] also proposed improvements for p-map analysis [1] by applying the Radon transformation on a pseudo p-map and looking for critical peaks in the transformed spectrum. As different resampling factors can lead to the same p-map and artefacts, it can be difficult to estimate an image's resampling factor. Pfennig and Kirchner [21] proposed to combine the analysis of p-maps with the spectral energy distribution to estimate the resampling factor of an image. Yet, image compression still introduces spurious spatial correlations that hinder the estimation.

Heuristic approaches other than p-map analysis were also proposed to investigate the local correlations in the spatial domain and reveal the periodic resampling artifacts in the Fourier domain. Gallagher [22] studied the hidden periodicity left by interpolation, and proposed to detect image resampling using the discrete Fourier transform (DFT) of the second difference. Following this idea, posterior works [23, 24] proposed to analyze the zero-crossings of the second difference of an image, and detect the image resampling in the DFT of the transformed image. Mahdian *et al.* [25] analyzed the hidden periodicity left by interpolation and proposed to detect traces of an affine transformation. Specifically, the Radon transform is employed to detect rotation traces. Then, resampling traces and noise analysis are combined for forgery detection [26]. Wang *et al.* [27] noticed that the local dependency between pixels induced by resampling leads to a lower rank of the matrix of a resampled image, and proposed to analyze the number of zero singular values of the singular value decomposition (SVD) for resampling detection. Building on this idea, Vázquez-Padín *et al.* [28] proposed a detection method that computes the SVD of a given image block and a measure of its degree of saturated pixels per row or per column, for discerning upsampled images from genuine ones. Vázquez-Padín *et al.* [29] presented an approach to design the optimal prefilter specific to a particular resampling factor and interpolation method for better estimating the resampling factor. Liu *et al.* [30] found that adjacent extrema intervals of a difference image obey a geometric distribution and that periodic peaks appear in the histogram of resampled JPEG images. They combined the characteristics of adjacent extrema intervals and spectrum artifacts to estimate downsampling factors on pre-JPEG-compressed images. Zhang *et al.* [31] exploited octant symmetric aliasing peaks (OSAP) caused by the coupling of resampling and post-JPEG compression to estimate resampling factors, and used an approximate OSAP model and a heuristic approach to enhance resampling features under low-quality JPEG compression.

Supervised learning-based methods have also been used to detect image resampling. A widely used approach is to leverage support vector machines (SVM) on hand-crafted features [32–34]. Methods based on deep neural networks have also been widely studied. Liang *et al.* [35] proposed a Convolutional Neural Network (CNN) to extract resampling feature patterns for detection. Bunk *et al.* [36] proposed a long-short-term memory (LSTM) network to detect and localize resampling traces. Flenner *et al.* [37] improved the method by Bunk *et al.* [36] by adopting the *a-contrario* framework to localize forged areas from resampling heat maps. Su *et al.* [38] modeled the resampled image as the result of a convolution between the original image and an unknown filter. They proposed to use a neural network for blind deconvolution, to simultaneously detect resampling and estimate the interpolation kernel. Lamba *et al.* [39] proposed an iterative pooling network and a branched network to tackle the resampling detection for images of varying dimensions. Cao *et al.* [40] presented a convolutional neural network to extract the resampling traces left in primary JPEG images followed by post-JPEG compression, and considered the resampling detection task as a multi-class classification problem. Bayar *et al.* [41] designed a constrained CNN to learn the low-level forensics traces and suppress image contents for detecting resampling in re-compressed images. Chen *et al.* [42] employed a densely connected CNN optimized for forensics to detect image resampling as well as other types of image manipulation. All of these methods show their effectiveness in resampling detection on several datasets. Nevertheless, supervised methods are not explainable, so no diagnosis is associated with a detection. Moreover, they suffer from domain-dependency and their generalization to new images with different statistics cannot be guaranteed.

In this paper, we point out that instead of analyzing the correlations between pixels in the spatial domain like in [22, 24, 25, 43], the resampling detection problem can be addressed by investigating spectral correlations in the Fourier domain. It is well known that sampling a signal with a sampling rate inferior to the Nyquist frequency folds the frequency components above the Nyquist frequency onto the low frequencies and causes aliasing [44]. A similar phenomenon occurs in image resampling. When an image is downsampled, the high-frequency components can be folded onto the low frequencies unless a "perfect" anti-aliasing filter, removing all the aliased frequencies, is applied prior to resampling. A "perfect" anti-aliasing via a brutal cut-off of all the high frequencies above the Nyquist frequency leaves undesirable ringing effects in the edges of the image. A more practical anti-aliasing is to apply a smooth cut-off of the frequencies near the band-limit to balance the reduction between aliasing artifacts and ringing artifacts, which unavoidably leaves folding traces in the spectrum. As for upsampling, except for the upsampling by the sinc interpolation (which is equivalent to zero padding in the Fourier domain) we shall see that upsampling by any other interpolation method inevitably replicates the low frequencies to the high frequency band (see Fig. 1). In general, one can show that the resampling operation periodizes and folds the original spectrum to the destination spectrum, leaving anomalous correlations between the replicated frequency components in the destination spectrum.

We therefore propose to use the spectral correlation in a resampled image to detect anomalous correlations between spectrum patches. Furthermore, we shall see that the detected correlations are very unlikely in the residuals of non-resampled images. Hence, we can evaluate their significance using the *a contrario* statistical framework. This enables the computation of unsupervised detection thresholds. A non-linear residual extraction filter is adopted before analysis. It enables suppressing the image content, which may cause false positive detections, and uncovers the subtle resampling traces. Extensive experiments show that the proposed resampling detection method is effective in different resampling scenarios and that it considerably outperforms all previous detection methods.

Our contributions are as follows:

- We conduct a theoretical study of the effect in the Fourier domain of signal resampling by interpolation. We specifically derive general formulas that reveal the periodization and folding of the spectrum.
- We establish the existence of spectral correlations caused by the spectrum folding phenomenon that occurs during resampling. Furthermore, we show that these are detectable and can be used as a trace of resampling.
- We present a method based on the *a contrario* framework that provides unsupervised detection thresholds to statistically validate the detected anomalous spectral correlations in the residual spectrum.
- A detailed comparison demonstrates that the method outperforms the compared methods by a large margin.

II. Fourier Analysis of Resampling

For the sake of clarity, we detail our notation for the discrete Fourier transform. We set $\omega_N := e^{\frac{i2\pi}{N}}$, $N \in \mathbb{R}_+$. The Discrete Fourier Transform (DFT) transforms a sequence of N samples $\{x_k\} := x_0, ..., x_{N-1}$ into another sequence of N complex numbers $\{\tilde{x}_n\} := \tilde{x}_0, ..., \tilde{x}_{N-1}$ by

Symbol	Description
u	an original signal in the spatial domain
$\tilde{u}$	the DFT of an original signal
v	a resampled signal in the spatial domain
$\tilde{v}$	the DFT of a resampled signal
φ	an interpolating filter
$c_n(\varphi)$	the n-th Fourier coefficient of an interpolating filter φ
M	the size of an original signal
N	the size of a resampled signal

TABLE I: Table of notations used in Sec. II.

$$\tilde{x}_n = \frac{1}{N}\sum_{k=0}^{N-1} x_k \cdot \omega_N^{-nk}, \qquad n = 0, \dots, N-1. \tag{1}$$

The sequence $\{\tilde{x}_n\}$ defined above for $n = 0, \dots, N-1$ can be extended to $\mathbb{Z}$. The resulting sequence shall have a period of N since $\{\tilde{x}_n\}$ is N-periodic by definition. The inverse Discrete Fourier Transform (IDFT) transforms $\{\tilde{x}_n\}$ back to $\{x_k\}$ by

$$x_k = \sum_{n=0}^{N-1} \tilde{x}_n \cdot \omega_N^{kn}, \qquad k = 0, \dots, N-1. \tag{2}$$

The Fourier series of an arbitrary real-valued T-periodic function $f \in L^2(0,T)$ is given by

$$f(x) = \sum_{n\in\mathbb{Z}} c_n(f) \cdot \omega_T^{nx}, \qquad x \in \mathbb{R} \tag{3}$$

where $c_n(f) = \frac{1}{T}\int_0^T f(x)\cdot\omega_T^{-nx}\,dx$ are the Fourier coefficients associated with f. The notation shared by the following sections is summarized in Tab. I.

A. Spectrum folding formulas for resampling methods

Consider a 1D signal $\{u_k\} \in \mathbb{R}^M$ of M samples and its resampled version $\{v_k\} \in \mathbb{R}^N$ of N samples. Here, $M, N \in \mathbb{N}^*$ can be arbitrary positive integers. All resampling methods proceed in two steps: first, the samples $\{u_k\}$ are interpolated to a continuous function f using an interpolating filter φ, then f is discretized in a signal $\{v_k\}$ with N samples:

$$\{u_k\} \in \mathbb{R}^M \xrightarrow[\varphi]{\text{interpolation}} f \xrightarrow{\text{discretization}} \{v_k\} \in \mathbb{R}^N.$$

The interpolation of $\{u_k\}$ to a continuous function $f : \mathbb{R} \to \mathbb{R}$ is done by convolving $\{u_k\}$ with an interpolating filter φ:

$$f(x) = \sum_{k=0}^{M-1} u_k \cdot \varphi(x-k), \qquad x \in \mathbb{R}. \tag{4}$$

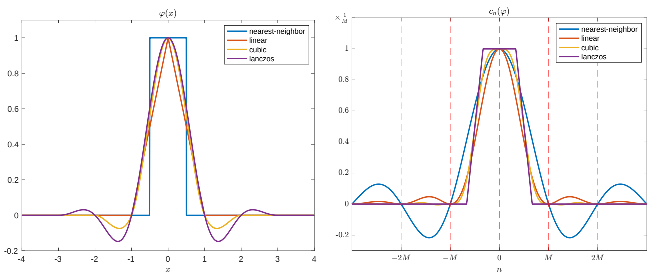
Common interpolating filters φ include the nearest-neighbor, bilinear, bicubic and Lanczos filters. Their expressions and Fourier coefficients $c_n(\varphi)$ are summarized in Tab. II and visualized in Fig. 3.

Then the resampling from f to $\{v_k\}$ is achieved by evenly discretizing f on $[0, M]$ to N samples:

$$v_k = f\left(k\frac{M}{N}\right), \qquad k = 0, \dots, N-1. \tag{5}$$

Interpolator	Expression of $\varphi(x)$ defined on $[-\frac{M}{2}, \frac{M}{2}]$	Fourier coefficients $c_n(\varphi)$
Nearest neighbor	$\begin{cases} 1, & \text{if } \lvert x\rvert \le \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{M}\,\mathrm{sinc}(\frac{n}{M})$
Linear	$\begin{cases} 1-\lvert x\rvert, & \text{if } \lvert x\rvert \le 1 \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{M}\,\mathrm{sinc}^2(\frac{n}{M})$
Cubic	$\begin{cases} \frac{3}{2}\lvert x\rvert^3 - \frac{5}{2}x^2 + 1, & \text{if } \lvert x\rvert \le 1 \\ -\frac{1}{2}\lvert x\rvert^3 + \frac{5}{2}x^2 - 4\lvert x\rvert + 2, & \text{if } 1 < \lvert x\rvert \le 2 \\ 0, & \text{otherwise} \end{cases}$	$\frac{1}{M}\left(3\mathrm{sinc}^4\left(\frac{n}{M}\right) - 2\mathrm{sinc}^2\left(\frac{n}{M}\right)\mathrm{sinc}\left(\frac{2n}{M}\right)\right)$
Lanczos	$\begin{cases} \mathrm{sinc}\,(x)\,\mathrm{sinc}\left(\frac{x}{3}\right), & \text{if } \lvert x\rvert \le 3 \\ 0, & \text{otherwise} \end{cases}$	$\frac{18}{M}\,\left(\mathrm{Rect} * \mathrm{Rect}(3\,\cdot) * \mathrm{sinc}(6\,\cdot)\right)\left(\frac{n}{M}\right), \mathrm{Rect}(\xi) := \begin{cases} 1, & \text{if } \lvert\xi\rvert \le \frac{1}{2} \\ 0, & \text{otherwise} \end{cases}$

TABLE II: The expressions and the Fourier coefficients of different M-periodic interpolating filters φ. $*$ is the discrete convolution of two Fourier series.



(a) φ (b) $c_n(\varphi)$

Fig. 3: The shapes of different interpolating filters φ and their Fourier coefficients $c_n(\varphi)$ listed in Tab. II.

Theorem 1 (Modulated Spectrum Folding in 1D)**.** *Consider an M-periodic interpolating filter φ. Denote the DFTs of $\{u_k\}$ and $\{v_k\}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier coefficients of φ by $\{c_p(\varphi)\}_{p\in\mathbb{Z}}$. Then we have*

$$\tilde{v}_n = M \cdot \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n}. \tag{6}$$

See Appendix A for the proof.

Theorem 1 describes the spectrum folding phenomenon in the Fourier domain when resampling a 1D signal of M samples into N samples with an interpolating filter. In simple words, it amounts to an N-periodization of the original M-periodic spectrum, modulated by the slow-varying Fourier coefficients of the interpolating filter.

The fact that the Fourier spectrum of the interpolating filter is slow-varying is essential, as it implies that one can observe near-replicas (up to a multiplicative constant) of the original spectrum in the spectrum of the interpolated image. The above theorem can be extended to any dimension. Here, we are particularly interested in 2D image resampling. Consider an image $\mathbf{u} \in \mathbb{R}^{M_1\times M_2}$, resampled to an image $\mathbf{v} \in \mathbb{R}^{N_1\times N_2}$ using an interpolating filter φ. Suppose that φ is (M_1, M_2)-periodic and that $\mathbf{u}$ is interpolated to an (M_1, M_2)-periodic function $f : \mathbb{R}^2 \to \mathbb{R}^2$. We obtain $\mathbf{v}$ by evenly sampling f on $[0, M_1] \times [0, M_2]$. The following Theorem again describes a spectrum folding phenomenon but in 2D.

Theorem 2 (Modulated Spectrum Folding in 2D)**.** *Denote the DFTs of $\mathbf{u}$ and $\mathbf{v}$ by $\{\tilde{u}_{m_1,m_2}\} \in \mathbb{C}^{M_1\times M_2}$ and $\{\tilde{v}_{n_1,n_2}\} \in \mathbb{C}^{N_1\times N_2}$, respectively. Denote the Fourier coefficients of φ and f by $\{c_{p_1,p_2}(\varphi)\}_{(p_1,p_2)\in\mathbb{Z}^2}$ and $\{c_{p_1,p_2}(f)\}_{(p_1,p_2)\in\mathbb{Z}^2}$, respectively. Then we have*

$$\tilde{v}_{n_1,n_2} = M_1 M_2 \sum_{z_1,z_2\in\mathbb{Z}} c_{z_1N_1+n_1, z_2N_2+n_2}(\varphi) \cdot \tilde{u}_{z_1N_1+n_1, z_2N_2+n_2}. \tag{7}$$

See Appendix B for the proof in any dimension.

B. Correlation between components in the spectrum

The Spectrum Folding Theorem 1 states that a frequency component in the output spectrum is composed of several frequency components in the original spectrum. Conversely, we can show that a frequency component in the original spectrum contributes to different frequency components in the output spectrum, making these frequency components correlated. The distance between two correlated frequency components in the output spectrum depends on the original and output sizes.

For simplicity, let us again analyze the 1D case, in which an original signal is resampled from M to N samples by an interpolating filter φ. As before, we denote the DFT of the original signal by $\{\tilde{u}_n\} \in \mathbb{C}^M$ and the DFT of the resampled signal by $\{\tilde{v}_n\} \in \mathbb{C}^N$. Denote the Fourier coefficients of the interpolating filter φ by $\{c_p(\varphi)\}_{p\in\mathbb{Z}}$.

From Theorem 1 we know that two frequency components separated by kM components, $k \in \mathbb{Z}$ being an arbitrary integer, can be correlated. Indeed, for any $n \in \mathbb{Z}$,

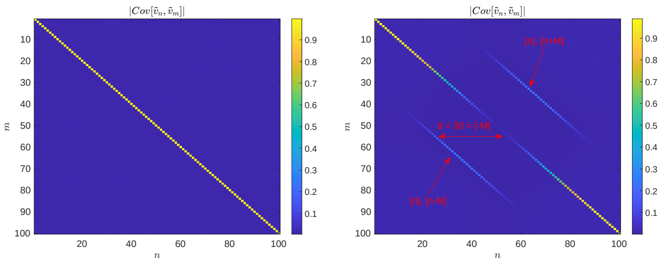
$$\tilde{v}_n = M \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \tag{8}$$

and as $\{\tilde{u}_m\}$ is M-periodic,

$$\begin{aligned} \tilde{v}_{n+kM} &= M \sum_{z\in\mathbb{Z}} c_{zN+n+kM}(\varphi) \cdot \tilde{u}_{zN+n+kM} \\ &= M \sum_{z\in\mathbb{Z}} c_{zN+n+kM}(\varphi) \cdot \tilde{u}_{zN+n} \end{aligned} \tag{9}$$

where we can see $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ are both weighted sums of the same original frequency components $\{\tilde{u}_{zN+n}, z \in \mathbb{Z}\}$ with different weights from $\{c_p(\varphi)\}$. Hence, $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ can be correlated due to the presence of the same $\{\tilde{u}_{zN+n}, z \in \mathbb{Z}\}$. We will study the properties of frequency correlation in the specific case of white noise, in particular, we shall see that:

- $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ are likely to be correlated for $k \in \mathbb{Z}^*$;



(a) Unaltered (b) Resampled

Fig. 4: $|\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m]|$ of the spectrum $\tilde{v}$ of (a) an unaltered white noise and (b) a resampled white noise. Both sequences have $N = 100$ samples. The original size of the resampled noise in (b) is $M = 70$. Note that two frequency components $\tilde{v}_n$ and $\tilde{v}_m$ in the spectrum of the resampled noise are likely to have strong covariance when $m = n \pm M(\mathrm{mod} N)$. On the other hand, any two different frequency components in the unaltered noise are uncorrelated.

- the correlation between $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ is the strongest when $|k| = 1$ and vanishes as $|k|$ increases within $[\![-\frac{1}{2}\mathrm{gcd}(M, N), \frac{1}{2}\mathrm{gcd}(M, N)]\!]$, where gcd denotes the *greatest common divisor*.

Fig. 4 shows the magnitude of the covariances of the DFTs $|\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m]|$ of an unaltered white noise and a resampled white noise, estimated from 10K simulations. Both noise sequences have $N = 100$ samples. The original size of the resampled white noise is $M = 70$. One can clearly see that two different frequency components $\tilde{v}_n$ and $\tilde{v}_m$ of the resampled noise are likely to be correlated when $m = n \pm M(\mathrm{mod} N)$, and are almost uncorrelated for other component pairs. Indeed, $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]$ is non-zero for any $k \in \mathbb{Z}$, but vanishes rapidly and is deemed negligible as $|k|$ increases. In contrast, any two different frequency components are uncorrelated for the unaltered white noise.

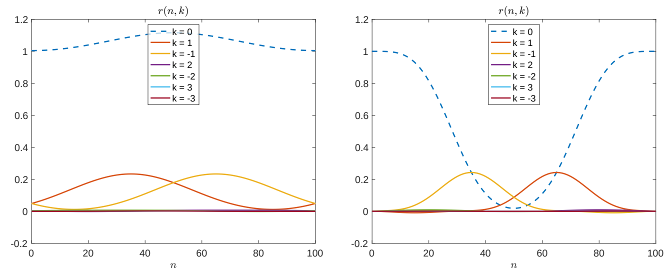
Theorem 3. *Let the original signal $\{u_k\} \in \mathbb{R}^M$ be a sequence of white noise with $\mathrm{Var}[u_k] = \sigma^2$ and $\mathrm{E}[u_k] = 0$, which is resampled to $\{v_k\} \in \mathbb{R}^N$ using an M-periodic interpolating filter $\varphi : \mathbb{R} \mapsto \mathbb{R}$. Let $\tilde{v}_m$ and $\tilde{v}_n$ be two arbitrary frequency components in the DFT of the resampled signal, $m, n \in \mathbb{Z}$. Denote by $L = lcm(M, N)$ and $gcd(M, M)$ the least common multiple and the greatest common divisor of M and N, respectively. We have $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m]$ given by:*

- *if $m-n$ is not a multiple of $\mathrm{gcd}(M, N)$, $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] = 0$;*
- *if $m - n$ is a multiple of $\mathrm{gcd}(M, N)$, the covariance is given by*

$$\begin{aligned}&\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] = \mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}] \\ &= M\sigma^2 \sum_{q\in\mathbb{Z}} \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM+qL}(\varphi)}\end{aligned} \tag{10}$$

where $k = \mathcal{K}(m - n)$ and $\mathcal{K}(x)$ is defined as the solution of

$$\begin{aligned}\min_{k'} \quad & |k'| \\ \textit{s.t.} \quad & k'M(\mathrm{mod} N) = x(\mathrm{mod} N). \\ & k' \in \mathbb{Z}\end{aligned} \tag{11}$$



(a) $M = 130$ and $N = 100$ (b) $M = 70$ and $N = 100$

Fig. 5: $n \mapsto r(n, k)$ for different small values of k in Eq. 12 for downsampling (Fig. 5a) and upsampling (Fig. 5b). $r(n, k)$ is non-negligible only for $k = 0, \pm 1$ and vanishes rapidly as $|k|$ increases.

See Appendix C for the proof.

Theorem 3 describes the covariance between two frequency components $\tilde{v}_n$ and $\tilde{v}_m$ of a resampled white noise, $m-n$ being divisible by $\mathrm{gcd}(M, N)$, as the sum of inner products of pairs of frequency components separated by $kM + qL$ components, where q traverses $\mathbb{Z}$ and $k = \mathcal{K}(m - n)$ is fixed. If we assume that the interpolating filter φ is a low-pass filter, its Fourier coefficients $c_p(\varphi)$ vanish as $|p|$ increases, and each inner product of Eq. S33 vanishes rapidly as the distance $|kM + qL|$ increases. The distance is independent of z and changes as q changes. Intuitively, we can expect the outer summation with index variable q to be dominated by the largest term associated to the dominant q of which the distance $|kM + qL|$ is minimal. Because the solutions of k' of Eq. S34 are $\mathbf{K} = \{k+q\frac{L}{M}, q \in \mathbb{Z}\}$ and by definition $|k| = \min_{k'\in\mathbf{K}} |k'|$, the minimal distance among $\{|kM + qL|, q \in \mathbb{Z}\}$ is $|kM|$. Therefore, the outer summation is dominated by the term at $q = 0$ and $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}] \approx M\sigma^2 \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi)\overline{c_{zN+n+kM}(\varphi)}$ where all the inner products share the common distance $|kM|$. More importantly, when n is fixed, if m changes so that $|k| = |\mathcal{K}(m - n)|$ increases, then the common distance $|kM|$ increases and $|\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m]|$ decreases.

To better illustrate the relationship between $|k|$ and $|\mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]|$, we define the normalized covariance at offset n and at order k as

$$\begin{aligned}r(n, k) :&= \frac{\mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]}{M\sigma^2} \\ &= \sum_{q\in\mathbb{Z}} \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM+qL}(\varphi)},\end{aligned} \tag{12}$$

which is real-valued for a real-valued and even interpolating filter[1] φ, and plot the curves of $n \mapsto r(n, k)$ corresponding to the cubic filter in the case of downsampling ($M > N$) or upsampling ($M < N$) for different small values of k in Fig. 5. It can be seen that $r(n, k)$ vanishes rapidly as $|k|$ increases, and is non-negligible only for $k = 0, \pm 1$. This confirms the observation in Fig. 4 that only the values on the upper and lower diagonals, $\{\mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n\pm M(\text{ mod } N)}]\}_n$, are non-negligible. The non-negligible correlation between $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ for small

[1]This is due to the fact that the Fourier coefficients of a real-valued and even function are real-valued.

$|k|$, especially for $|k| = 1$, is an important trace left by the resampling operation and hence a hint for its detection.

C. Correlation between patches in the spectrum

Despite the non-negligible covariance between two frequency components $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ for small $|k|$, in practice we only have access to one single sample of $\tilde{v}_n$ and $\tilde{v}_{n+kM}$ from the DFT of the signal in question, which makes the estimation of $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}]$ imprecise. We can instead measure the correlation between the means of two small segments of the spectrum separated by kM frequency components by varying n in a small interval. Concretely, let $\mathcal{I}_l = [\![-l, l]\!]$ be a small interval with $l \ll M$ and let $\tilde{w}_n = \frac{1}{l}\sum_{\Delta\in\mathcal{I}_l}\tilde{v}_{n+\Delta}$ and $\tilde{w}_{n+kM} = \frac{1}{l}\sum_{\Delta\in\mathcal{I}_l}\tilde{v}_{n+kM+\Delta}$ be the means of two segments separated by kM components, then the correlation between $\tilde{w}_n$ and $\tilde{w}_{n+kM}$ is given by

$$\begin{aligned}\rho(\tilde{w}_n, \tilde{w}_{n+kM}) &= \frac{|\mathrm{Cov}[\tilde{w}_n, \tilde{w}_{n+kM}]|}{\sqrt{|\mathrm{Var}[\tilde{w}_n]|}\sqrt{|\mathrm{Var}[\tilde{w}_{n+kM}]|}} \\ &\approx \frac{|\sum_{\Delta\in\mathcal{I}_l}\mathrm{Cov}[\tilde{v}_{n+\Delta}, \tilde{v}_{n+kM+\Delta}]|}{\sqrt{|\sum_{\Delta\in\mathcal{I}_l}\mathrm{Var}[\tilde{v}_{n+\Delta}]|}\sqrt{|\sum_{\Delta\in\mathcal{I}_l}\mathrm{Var}[\tilde{v}_{n+kM+\Delta}]|}}\end{aligned} \tag{13}$$

which amounts to calculating the Pearson correlation coefficient (PCC) of the two segments $\{\tilde{v}_{n+\Delta}\}_{\Delta\in\mathcal{I}}$ and $\{\tilde{v}_{n+kM+\Delta}\}_{\Delta\in\mathcal{I}}$ in practice. $\mathrm{Cov}[\tilde{w}_n, \tilde{w}_{n+kM}]$, $\mathrm{Var}[\tilde{w}_n]$ and $\mathrm{Var}[\tilde{w}_{n+kM}]$ in Eq. 13 is respectively expanded at the second equality as the sum of the covariances of components at distance kM and the variances of each of the component, because any other pairs of components with the two small segments separated at distances other than 0 or kM have negligible covariances.[2] Combining with Eq. 12, Eq. 13 can be further written as

$$\begin{aligned}&\rho(\tilde{w}_n, \tilde{w}_{n+kM}) \\ &\approx \frac{|\frac{1}{l}\sum_{\Delta\in\mathcal{I}_l} r(n+\Delta, k)|}{\sqrt{|\frac{1}{l}\sum_{\Delta\in\mathcal{I}_l} r(n+\Delta, 0)|}\sqrt{|\frac{1}{l}\sum_{\Delta\in\mathcal{I}_l} r(n+kM+\Delta, 0)|}}.\end{aligned} \tag{14}$$

The denominator is bounded as $n \mapsto r(n, 0)$ is bounded, and the numerator as a moving average of $n \mapsto r(n, k)$ is non-negligible. Thus, $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ is non-negligible. Since $\tilde{w}_n$ and $\tilde{w}_{n+kM}$ are the means of multiple frequency components, the correlation $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ is observable with less uncertainty. Also, the correlation eliminates the constant $M\sigma^2$ in the covariance and makes the measure invariant to the size and variance of the original signal.

Fig. 6. shows the correlation between two different segments $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ (Eq. 14) as a function of the offset n for different small values of k in the case of downsampled and upsampled white noise. Similarly to Fig. 5, we can see the correlation between two segments vanishes as $|k|$ increases and is the strongest at $|k| = 1$.

The same remark is valid in 2D. Consider a 2D signal of size $N_1 \times N_2$ resampled from an original 2D white noise of size $M_1 \times M_2$, and denote the DFT of the resampled signal by $\tilde{v}$. Let $k_1, k_2 \in \{\pm 1\}$. Let $\mathcal{I}_l = [\![-l, l]\!]$ be a small interval. Then the means of two small patches separated by (k_1M_1, k_2M_2) components,

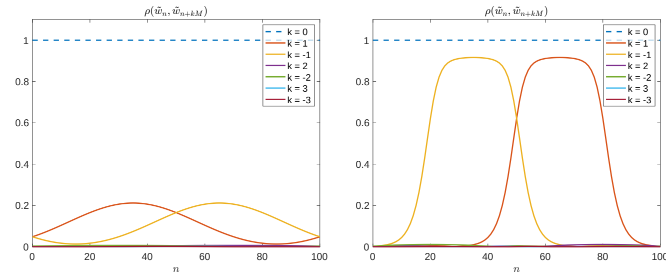
$$\tilde{w}_{n_1,n_2} = \frac{1}{l^2}\sum_{(\Delta_1,\Delta_2)\in\mathcal{I}_l^2}\tilde{v}_{n_1+\Delta_1,n_2+\Delta_2}$$

and

$$\tilde{w}_{n_1+k_1M_1,n_2+k_2M_2} = \frac{1}{l^2}\sum_{(\Delta_1,\Delta_2)\in\mathcal{I}_l^2}\tilde{v}_{n_1+k_1M_1+\Delta_1,n_2+k_2M_2+\Delta_2}$$

are highly correlated for most (n_1, n_2), which can be measured by calculating the PCC of the two patches.

We illustrate the correlations in the spectrum of a resampled image in Fig. 1, where the image has been resampled from 512×512 to 665×665. After residual extraction, two patches in the spectrum separated by $153 = -512 \pmod{665}$ components along the height have similar patterns, showing their strong correlation in between.



(a) $M = 130$ and $N = 100$ (b) $M = 70$ and $N = 100$

Fig. 6: $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ for different small values of k in Eq. 14 in the case of downsampling (Fig. 6a) and upsampling (Fig. 6b). $\rho(\tilde{w}_n, \tilde{w}_{n+kM})$ is non-negligible only for $k = 0, \pm 1$ and vanishes rapidly as $|k|$ increases. The length of the segments is 11.

III. *A contrario* DETECTION OF IMAGE RESAMPLING

In this section, we propose an *a contrario* method to detect image resampling by analyzing the anomalous correlations within the spectrum caused by resampling, as pointed out in Sec. II. Given an image under scrutiny, we first extract a residual from it with a non-linear filter, which aims at removing all natural structures and keeping only a residual, see Fig. 7. As we will see, the residual signal usually has statistics similar to white noise. Then, we detect anomalous correlations between patches of the residual spectrum.

A. Residual extraction

In Sec. II-C we showed that, for white noise, non-negligible spectral correlations are present when resampling. However, natural images differ largely from white noise; therefore, the conclusion of spectral correlations does not necessarily hold. In addition, the scene content in natural images can also introduce correlations between patch pairs and cause false positive detections (see Fig. 12).

To enhance the detectability of spectral correlations induced by resampling, residual extraction is performed prior to the

[2] $\mathrm{Cov}[\tilde{w}_n, \tilde{w}_{n+kM}] = \frac{1}{l^2}(\sum_{\Delta\in\mathcal{I}}\mathrm{Cov}[\tilde{v}_{n+\Delta}, \tilde{v}_{n+kM+\Delta}] + \sum_{\Delta_1,\Delta_2\in\mathcal{I}_l,\Delta_1\neq\Delta_2}\mathrm{Cov}[\tilde{v}_{n+\Delta_1}, \tilde{v}_{n+kM+\Delta_2}])$. When $|k|$ is small and $\Delta_2 - \Delta_1$ is small but non-zero, $(n+kM+\Delta_2)-(n+\Delta_1) = kM+\Delta_2-\Delta_1$ is either not a multiple of $\gcd(M, N)$ or related to $|\mathcal{K}(kM+\Delta_2-\Delta_1)| \gg 1$, which results in negligible $\mathrm{Cov}[\tilde{v}_{n+\Delta_1}, \tilde{v}_{n+kM+\Delta_2}]$.

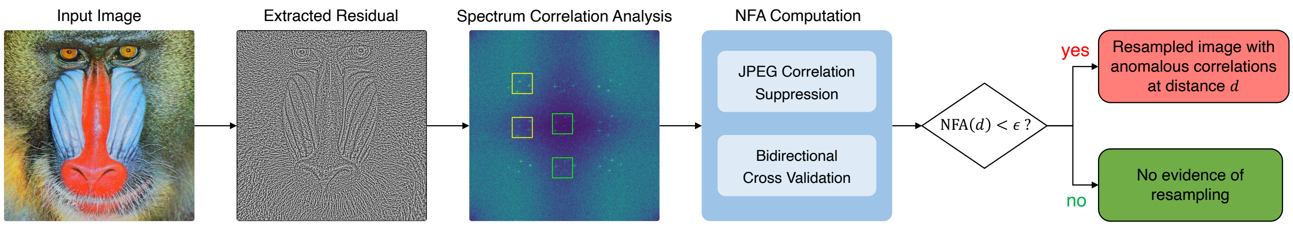


Fig. 7: The pipeline of the *a contrario* detection of image resampling by detecting anomalous correlations in the spectrum at different distances d.

detection to transform a natural image into a residual image. This step aims to approximate the distribution of a resampled white noise, thereby ensuring the validity of the theoretical analysis established in Sec. II-C. The extraction filter must effectively suppress structural scene content while preserving subtle resampling traces. Moreover, it should homogenize the spectrum such that it resembles a spectrum of resampled white noise and does not introduce additional correlations. To this end, two extraction filters are particularly well-suited for homogenizing the spectrum and mitigating false alarms:

- **Rank transform** [45] The rank transform R of an image I computes the rank of the intensity of each pixel x in its 7×7 neighborhood $\mathcal{N}(x)$
$$R(x) = \sum_{x' \in \mathcal{N}(x)} \mathbb{1}_{\{I(x') < I(x)\}}, \tag{15}$$
which preserves local structures while neglecting long-range variations.
- **Total variation (TV) denoiser** [46, 47] The TV denoiser extracts the bounded variation component of each image, which is piecewise smooth with sharp discontinuities. The image residual contains the image noise and the tiny oscillations caused by resampling.

Usually the rank transform is more suitable for uncompressed images, whereas the TV denoiser is more suitable for JPEG-compressed images. This will be further evaluated quantitatively in Sec. IV-E and qualitatively in Sec. IV-I.

B. A contrario detection of anomalous local correlations

Suppose that an image of size $M_1 \times M_2$ is resampled to an image of size $N_1 \times N_2$. In Sec. II-B, we have shown that small patches of the spectrum separated by $k_1 M_1$ and $k_2 M_2$ frequency components along the first and second axis are correlated, where k_1 and k_2 are two arbitrary integers, and the correlation vanishes as $|k_1|$ or $|k_2|$ increases and becomes too weak to be detected. In the following, we shall describe the resampling detection along one axis, being the application to the other axis analogous. Taking into consideration the period (N_1, N_2) of the spectrum in scrutiny, we therefore aim at detecting correlated patches that are separated by $\pm M_1 (\mathrm{mod}\, N_1)$ or $\pm M_2 (\mathrm{mod}\, N_2)$ frequency components along the first or second axis, respectively.

A detection threshold is necessary to point out the anomalous local correlations in the spectrum of a resampled image. Such thresholds will be obtained by the non-parametric *a contrario* framework. The *a contrario* framework is based on a general perception principle, the Helmholtz principle [48], which states that a structure is perceived whenever some large deviation from randomness occurs. The structure to perceive is a pair of correlated spectrum patches separated by $d = M_1 (\mathrm{mod}\, N_1)$ or $d = -M_1 (\mathrm{mod}\, N_1)$ frequency components along the axis.

The proposed *a contrario* method is based on the null hypothesis that the image has not been resampled. In the residual of such an image, given an arbitrary distance d, there is no particular reason to observe a large number of correlated patch components. Thus, if we observe a large number of such correlated pairs of spectrum patches, the image is very likely to have been resampled. The linear correlation between spectrum patches is measured by the Pearson Correlation Coefficient (PCC). Since the spectrum patches are complex-valued, the PCC should also be defined accordingly. Let $\mathbf{x}, \mathbf{y} \in \mathbb{C}^N$ be two vectorized local patches, the Pearson correlation coefficient between $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathrm{corr}(\mathbf{x}, \mathbf{y}) := \left| \frac{\langle \mathbf{x} - m(\mathbf{x}), \mathbf{y} - m(\mathbf{y}) \rangle}{\|\mathbf{x} - m(\mathbf{x})\|_2 \, \|\mathbf{y} - m(\mathbf{y})\|_2} \right|, \tag{16}$$

where $m(\cdot)$ denotes the mean of a vector, $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=0}^{N-1} x_i \bar{y}_i$ the inner product between $\mathbf{x}$ and $\mathbf{y}$, and $\|\cdot\|_2$ the ℓ_2-norm of a vector.

We must now decide whether a measured PCC correlation between two patches is significantly large to be anomalous. This can be decided from the image spectrum itself by a non-parametric comparison between the measured correlation and the observed background correlations from other random pairs of patches.

Let us denote by $\tilde{v}$ the DFT of the tested image. Let $I_i = [\![n_1^i, n_1^i + l_1 - 1]\!] \times [\![n_2^i, n_2^i + l_2 - 1]\!]$ be a small patch of frequency entries, with $l_1 \ll N_1$ and $l_2 \ll N_2$. Let $P_i = (\tilde{v}_{n_1,n_2})_{(n_1,n_2) \in I_i}$ be an anchor patch of the spectrum, and $Q_i = (\tilde{v}_{n_1+d,n_2})_{(n_1,n_2) \in I}$ be the patch that is d components apart from P_i in the first axis. Denote by $\rho(P_i, d) := \mathrm{corr}(P_i, Q_i)$ the correlation of the anchor patch P w.r.t. another patch lying d coefficients apart. Under the null hypothesis H_0, the correlations for an interval of $2r+1$ distances, $\mathcal{N}_r(P_i, d) := \{\rho(P_i, d') : d' \in [\![d-r, d+r]\!]\}$, should have similar values. The alternative hypothesis is that the image has been resampled and that for the particular distance d, $\rho(P_i, d)$ is very likely to be a local maximum in $\mathcal{N}_r(P_i, d)$.

This can be tested by counting the number of maximum correlations among all patch pairs. To this end, we divide the domain of $\tilde{v}$ into small non-overlapping patches $\mathcal{P} = \{P_i\}$. The number of correlation maxima among all considered patch pairs is

$$k(d) := \sum_{P_i \in \mathcal{P}} \mathbb{1}_{\left\{\rho(P_i,d)=\max \mathcal{N}_r(P_i,d)\right\}} \tag{17}$$

where $\mathbb{1}_{\{\cdot\}}$ is the indicator function. If $k(d)$ is larger than a certain anomaly threshold κ, then we shall say that there is an unusual large number of correlated patch pairs at the distance d in the spectrum.

Instead of directly fixing the anomaly threshold κ, the *a contrario* framework controls the number of falsely detected distances d just by chance in a random spectrum. This depends on the number of tested distances. Since the spectrum is N_1-periodic along the axis of detection, it is enough to test all distances d satisfying $d < N_1$. In practice, we shall ignore the distance candidates that are too close to the spectrum end points, as they do not have enough neighboring distances to validate the maximum correlation. Therefore, we only test distance candidates in the restricted domain $\mathcal{C} = [\![r+1, N_1 - r - 1]\!]$.

Denote the random variable $k(d)$ by $\mathbf{k}(d)$. The *a contrario* framework implicitly sets κ such that the expected number of detections is smaller than a certain ϵ under H_0:

$$\kappa \in \{\eta : \ \mathbb{E}_{H_0}\big[\sum_{d\in\mathcal{C}} \mathbb{1}_{\mathbf{k}(d)\geq\eta}\big] = \#\mathcal{C}\cdot\mathbb{P}_{H_0}(\{\mathbf{k}(d)\geq\eta\}) < \epsilon\} \tag{18}$$

where $\#\mathcal{C} = N_1 - 2r - 1$ is the total number of tested distances. Taking the lower bound of the set of possible thresholds, κ is set to satisfy $\#\mathcal{C}\cdot\mathbb{P}_{H_0}(\mathbf{k}(d)\geq\kappa) = \epsilon$. We define the Number of False Alarms (NFA) of a tested distance d by

$$\text{NFA}(d) := \#\mathcal{C}\cdot\mathbb{P}_{H_0}\big(\mathbf{k}(d) > k(d)\big). \tag{19}$$

Then, verifying the threshold condition $k(d) > \kappa$ is equivalent to ensuring $\text{NFA}(d) < \epsilon$, which validates a detection with a false alarm rate smaller than ϵ.

It remains to calculate $\mathbb{P}_{H_0}\big(\mathbf{k}(d)\big)$. Under H_0, the probability that a correlation $\rho(P_i, d)$ is the maximum of $\mathcal{N}_r(P_i, d)$ is $\frac{1}{2r+1}$. Since the patches in $\mathcal{P}$ do not overlap, this ensures the independence of $\rho(P_i, d)$ for any two different P_i. Thus, under H_0, $\mathbf{k}(d)$ follows a binomial distribution

$$H_0 : \mathbf{k}(d) \sim B\left(\#\mathcal{P}, \frac{1}{2r+1}\right) \tag{20}$$

where $\#\mathcal{P}$ is the number of non-overlapped patches. Finally, a distance d is validated if

$$\text{NFA}(d) = (N_1 - 2r - 1)\cdot\mathbf{B}\left(k(d), \#\mathcal{P}, \frac{1}{2r+1}\right) < \epsilon \tag{21}$$

where $\mathbf{B}$ is the tail of the binomial law, $\mathbf{B}(k,n,p) = \sum_{i=k}^{n}\binom{n}{i}p^i(1-p)^{n-i}$. The NFA is computed for all distances $d \in \mathcal{C}$, and the image is classified as a resampled image if at least one distance d is validated.

It is worth noting that once an anomalous d is detected, the original size M_1 cannot be determined due to the ambiguity of M_1 in $d = \pm M_1(\bmod N_1)$, but d reduces the possible values of M_1 to a limited set $\{\pm d + kN_1, k\in\mathbb{Z}\}\cap\mathbb{N}_+^*$, which is an important clue of the original size.

C. Suppression of JPEG traces

It has been shown in previous works [22, 30, 31] that JPEG compression using 8×8 blocks leaves traces that are similar to a resampling of factor 8. This can be understood by considering the extreme case where JPEG compression only keeps the DC component (the mean) of each block. Hence the compressed image is the result of a) zooming down the image by an 8 factor using the 8×8 block filter, and b) zooming in the image by 8 using the nearest neighbor interpolation filter. It is easily checked that these operations entail high spectral correlations peaks at distances $d \in \{\frac{kN_1}{8}, k = 1, ..., 7\}$. Besides, one can observe that in JPEG images the correlations at distances close to $\{\frac{kN_1}{8}, k = 1, ..., 7\}$ are also slightly higher than other correlations. These high spectral correlations result in false detections, and also hinder the validation of any distance related to image resampling whose comparison interval contains the high JPEG correlations.

To suppress the spurious distances related to authentic JPEG compression, we need to exclude correlations in the neighborhoods centered around the distances related to JPEG compression. Specifically, for each anchor patch P_i and each distance $d^* \in \{\frac{kN_1}{8}, k = 1, \dots, 7\}$ related to JPEG compression, let d^- and d^+ be the second *local minimum* [3] of $\rho(P_i, \cdot)$ on either side of d^*, we discard the correlations $\rho(P_i, d)$ for all $d \in [\![d^-, d^+]\!]$. All these discarded correlations are considered invalid when counting the number of correlation maxima at distance d in Eq. 17: 1) a discarded $\rho(P_i, d)$ will not be counted as a maximum correlation; 2) the neighboring correlations are updated as $\mathcal{N}_r(P_i, d) := \{\rho(P_i, d') :$ *d' is among the r nearest valid distances on each side of d*$\}$ to ensure the fixed number of $2r+1$ correlations. An example of the suppression of JPEG distances on a non-resampled image with post-JPEG compression is shown in Fig. 8.

D. Cross validation for detecting proportional resampling

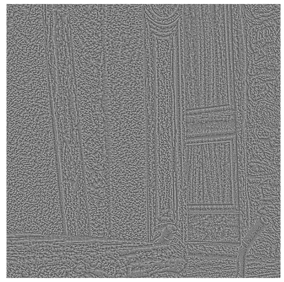
Image rescaling mostly occurs in equal proportions along the width and height, thus preserving the aspect ratio of the image. If we assume such proportional resampling, then spectral correlation distances are expected to be proportional in both directions. This enables to further eliminate false detections due to the image contents. Indeed, false positive distances caused by the image content are usually independent in both directions.

Consider an image of original size $M_1 \times M_2$, resampled at ratios q_1 and q_2 respectively along the two axes to size $N_1 \times N_2$, where $N_1 = q_1 M_1$ and $N_2 = q_2 M_2$. The expected positive distances along the first axis are $\mathcal{D}_1 = \{kM_1 \bmod N_1 = N_1(\frac{k}{q_1} - \lfloor\frac{k}{q_1}\rfloor), k \in \mathbb{Z}\}$, and along the second axis are $\mathcal{D}_2 = \{N_2(\frac{k}{q_2} - \lfloor\frac{k}{q_2}\rfloor), k\in\mathbb{Z}\}$. If resampling is performed in equal proportions along both axes, i.e., $q_1 = q_2 = q$, then the positive distances should share the same ratios $\{\frac{k}{q} - \lfloor\frac{k}{q}\rfloor, k\in\mathbb{Z}\}$ relative to N_1 and N_2, respectively. This property allows us to perform cross-validation between the positive distances along both axes.

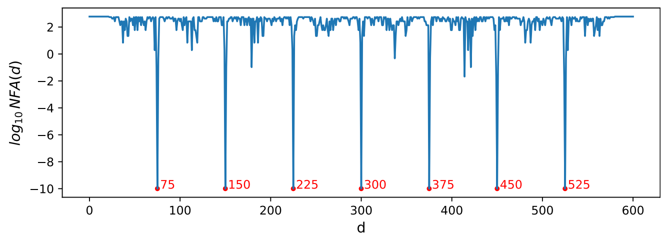
In practice, after computing the NFAs along both axes denoted as $\{\text{NFA}_1(d_1), d_1 < N_1\}$ and $\{\text{NFA}_2(d_2), d_2 < N_2\}$,

[3] Empirically the correlation $\rho(P_i, d^*)$ is never a local minimum of $\rho(P_i, \cdot)$.

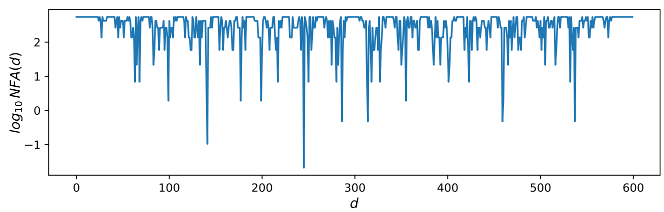
(a) original JPEG image



(b) residual image



(c) NFAs without suppression



(d) NFAs with suppression of distances close to JPEG distances

Fig. 8: Detection on a *non-resampled* image compressed by JPEG. Detection results present significant low NFAs in the neighborhoods of JPEG distances $\{\frac{kN_1}{8}, k = 1, ..., 7\}$ without suppression. These are successfully eliminated by our suppression strategy.

we validate a distance d_1 along the first axis as positive if the corresponding distance along the second axis, related by the same ratio, is also positive:

$$\mathrm{NFA}_1^{\mathrm{valid}}(d_1) = \max\left(\mathrm{NFA}_1(d_1), \mathrm{NFA}_2\left(\frac{d_1 N_2}{N_1}\right)\right). \quad (22)$$

To account for slight deviations from perfect proportional resampling, we allow a small relative tolerance $\beta \ll 1$ between q_1 and q_2, which is roughly equivalent to allowing the same tolerance between the ratios of the distances to the image sizes along both axes. The NFA after cross-validation on the first axis is finally computed as

$$\mathrm{NFA}_1^{\mathrm{valid}}(d_1) = \max\left(\mathrm{NFA}_1(d_1), \min_{\left|\frac{x}{N_2}-\frac{d_1}{N_1}\right|<\beta} \mathrm{NFA}_2(x)\right). \quad (23)$$

IV. Experiments

In this section, we present experimental results to validate the effectiveness of our proposed image resampling detection method. Our method is implemented using the hyperparameters summarized in Table III. Specifically, we employ the implementation of TV denoising from [47], and adapt the rank transform window size to account for JPEG compression artifacts. The *a contrario* detection is computationally optimized via image integral techniques [49] and the computation of correlations by FFT. Comprehensive details on the algorithm's implementation and computational complexity are provided in Appendix F.

Hyperparameter	**Value**
Rank transform window (PNG / JPEG)	7×7 / 3×3
TV regularization param.	1
Patch size	$H/10 \times W/10$
Radius of neighborhood $\mathcal{N}_r$	$r = 20$
Cross-validation tolerance	$\beta = 0.005$

TABLE III: Implementation parameters

Evaluation was conducted using several datasets: RAISE [50] was used for comparison against unsupervised methods and for the ablation study; BOSSBase [51] and Dresden [52] were employed for comparison against supervised methods; and a combination of RAISE [50] and FODB [53] was utilized for comparison in real-world scenarios.

A. Comparison with unsupervised methods

The first comparison aims to evaluate the performance of our unsupervised method against five existing unsupervised resampling detection methods in the literature: Popescu [1], Gallagher [22], Birajdar [24], Mahdian [25], and Qiao [43]. Experiments were carried out using 1000 uncompressed images from the Raise-1k dataset [50]. As the demosaicing traces are similar to upsampling traces by a factor 2, the images were first downsampled by a factor 2 to remove these traces. Then, these images were resampled at different factors and cropped to 600×600 in size, resulting in 1000 resampled images for each resampling factor. In addition, the same images without resampling were directly cropped to 600×600 to serve as original images. Experiments were performed on grayscale images, the extension to color being straightforward.

A.1. Without anti-aliasing We first evaluated the detection performance on resampled images without anti-aliasing filtering. Since all of the compared methods require a threshold to distinguish original and resampled images, we set a detection threshold ensuring a false positive rate of 1% for each method, and compared their detection accuracies.

The comparison was carried out on images resampled at factors between 0.6 and 1.6, as downsampled images at a factor below 0.5 are almost undetectable and upsampled images at a factor above 1.6 are easy to detect. No anti-aliasing was applied for downsampling. The resampled images were saved respectively in uncompressed PNG format and JPEG format at qualities 95 and 90. In the residual extraction step, our method uses the rank transform for PNG images and the TV denoiser for JPEG images. The performance of our method was evaluated on horizontal resampling detection, and also on resampling detection with bidirectional cross validation by assuming that the resampling ratio is the same in both horizontal and vertical dimensions.

The results are shown in Tab. IV. Our method outperforms all the compared methods for all the resampling factors and compression qualities. For uncompressed PNG images, our method without cross-validation attains an accuracy over 93% at resampling factor 0.6 and detects almost all the resampled

	Resampling factor	0.6	0.7	0.8	0.9	1.1	1.2	1.3	1.4	1.5	1.6	average
PNG	Qiao [43]	0.3	0.5	0.6	0.9	2.3	3.5	5.7	9.5	17.0	25.1	6.5
	Mahdian [25]	4.5	25.1	3.5	12.5	62.5	90.8	98.2	98.1	93.3	96.7	58.5
	Gallagher [22]	4.7	25.4	34.7	37.0	68.0	93.1	98.5	98.4	96.0	98.1	65.4
	Birajdar [24]	2.1	31.0	16.3	46.0	95.9	98.4	99.2	99.4	98.6	99.3	68.6
	Popescu [1]	30.0	77.3	73.9	1.3	16.8	97.5	98.9	99.1	98.7	98.8	69.2
	Ours	93.1	99.9	**100.0**	99.8	99.7	**99.9**	99.8	99.8	99.8	99.8	99.2
	Ours, cross-val.	**98.1**	**100.0**	99.9	**99.9**	**100.0**	**99.9**	**100.0**	**99.9**	**100.0**	**100.0**	**99.8**
JPEG Q=95	Qiao [43]	0.3	0.8	-	0.8	1.6	1.9	2.3	3.6	4.0	-	1.9
	Mahdian [25]	3.6	20.1	-	9.7	51.2	77.1	93.0	93.2	83.6	-	53.9
	Gallagher [22]	3.9	25.5	-	33.4	66.7	87.6	96.1	95.6	90.5	-	62.4
	Birajdar [24]	1.4	11.6	-	15.5	83.9	87.3	93.5	97.0	93.0	-	60.4
	Popescu [1]	0.0	0.6	-	1.1	1.3	2.6	1.3	2.4	5.7	-	1.9
	Ours	48.0	78.1	-	**93.0**	**98.0**	98.9	**97.5**	98.1	**99.6**	-	88.9
	Ours, cross-val.	**60.0**	**88.4**	-	**93.0**	97.5	**99.1**	**97.5**	**98.6**	**99.6**	-	**91.7**
JPEG Q=90	Qiao [43]	0.4	0.8	-	0.8	1.4	1.7	1.8	2.6	3.2	-	1.6
	Mahdian [25]	2.8	13.8	-	5.8	38.4	58.9	77.5	83.8	73.3	-	44.3
	Gallagher [22]	3.3	13.8	-	15.3	44.0	63.9	77.6	84.7	77.6	-	47.5
	Birajdar [24]	2.2	6.6	-	9.6	58.3	65.3	77.2	87.4	75.1	-	47.7
	Popescu [1]	0.0	0.0	-	0.3	0.7	0.1	0.0	0.3	0.2	-	0.2
	Ours	26.6	41.3	-	**66.8**	88.2	**93.1**	83.9	86.9	**96.5**	-	72.9
	Ours, cross-val.	**36.0**	**55.8**	-	66.1	**89.0**	93.0	**88.6**	**90.6**	96.1	-	**76.9**

TABLE IV: Detecting accuracy (%) of different detection methods at a 1% false positive rate and resampling factors ranging from 0.6 to 1.6. The resampled images were saved in PNG format and JPEG format at qualities 95 and 90. The image resampling at factors 0.8 and 1.6 was ignored because the correlation peaks due to resampling overlap with those caused by JPEG compression.

images at higher resampling factors, substantially ahead of all the compared methods. Using cross-validation to suppress false positives, the detection accuracy of the resampling factor 0.6 is increased to 98.1%. When the resampled images are saved in JPEG at quality factor 95, the detection accuracies are still higher than 97% for upsampled images, but drop significantly on downsampled images compared to uncompressed images. The cross-validation is helpful to boost the detection performance of downsampled images. At JPEG quality factor 90, the cross-validation only has limited effect on accuracy, because the resampling traces are severely altered by JPEG compression. Nevertheless, the performance of our method is still superior to the compared methods on strongly compressed images.

A.2. With anti-aliasing A low-pass filter is usually applied to the image before downsampling in order to prevent aliasing in the downsampled image. If an image has undergone a strictly band-limited filtering, no frequency should be folded in the downsampled image, and the image downsampling would no longer be detectable. Applying a strict band-limited anti-aliasing filter is equivalent to a convolution with a sinc filter in the spatial domain, which results in undesirable ringing effects. A smooth frequency cut-off is, therefore, usually preferred as a compromise between ringing and aliasing. Such a strategy entails some frequency folding that can still be detected by our method. Hence, two representative anti-aliasing strategies were considered to compare the performances of different methods: 1) anti-aliasing with Gaussian blur; 2) anti-aliasing by stretching the support of interpolation kernel.

Gaussian blur is widely used as anti-aliasing filter. Assuming that an initial optical Gaussian blur with standard deviation σ_a is sufficient to ensure an approximately well-sampled image, a relative blur with $\sigma = \sigma_a\sqrt{1/r^2 - 1}$ is required to maintain well-sampling for the resampled image by a factor $r < 1$. In our experiments, we conservatively assumed $\sigma_a = 0.8$ which yields a slightly blurry image, compared to the practical value of $\sigma_a = 0.5$ assumed for the SIFT keypoint extraction [54]. This allowed us to evaluate the downsampling detection performance after a very conservative anti-aliasing. This comparison was carried out on images resampled at resampling factors between 0.6 and 0.95 with the bicubic interpolator and saved in PNG. Its results

Factor r	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95	avg.
Qiao [43]	1.4	1.3	1.2	0.9	1.3	1.0	1.1	1.2	1.2
Mahdian [25]	47.6	32.5	1.1	6.4	2.4	3.7	11.2	3.3	13.5
Gallagher [22]	52.4	30.9	1.2	7.0	11.7	21.4	31.8	20.4	22.1
Birajdar [24]	43.3	57.9	22.0	39.1	10.0	15.0	37.6	75.5	37.6
Popescu [1]	60.6	12.5	1.3	2.1	0.8	0.5	1.1	0.8	10.0
Ours	88.0	91.1	81.1	97.0	95.8	95.7	99.7	**99.5**	93.5
Ours, c.v.	**96.5**	**97.9**	**94.5**	**99.5**	**98.9**	**97.4**	**99.8**	99.4	**98.0**

TABLE V: Detecting accuracy (%) of different detection methods at false positive rate of 1% on images downsampled with anti-aliasing by Gaussian blur ($\sigma = 0.8\sqrt{1/r^2 - 1}$) at different resampling factors r from 0.6 to 0.95, with bicubic interpolation.

Factor	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95	avg.
Qiao [43]	1.4	1.3	1.2	0.9	1.3	1.0	1.1	1.2	1.2
Mahdian [25]	1.7	22.2	6.1	18.1	1.9	7.6	9.9	3.6	8.9
Gallagher [22]	2.1	21.6	6.3	21.8	15.2	20.2	25.4	20.2	16.6
Birajdar [24]	3.1	3.6	5.7	12.4	3.5	23.0	56.8	85.0	24.1
Popescu [1]	5.5	**76.1**	21.7	**49.6**	29.5	25.8	6.0	2.4	27.1
Ours	**31.0**	35.5	**35.6**	41.1	66.8	86.4	96.4	97.5	**61.3**
Ours, c.v.	24.5	29.9	29.3	41.0	**73.3**	**89.1**	**96.7**	**98.1**	60.2

TABLE VI: Detecting accuracy (%) of different detection methods at false positive rate of 1% on images downsampled with anti-aliasing by spatial-domain kernel stretching (see Eq. 24) at resampling factors from 0.6 to 0.95, with bicubic interpolation.

are shown in Tab. V. As was expected, our detector performs generally worse on anti-aliased images than on aliased images. Surprisingly, four of the compared methods Mahdian [25], Gallagher [22], Birajdar [24] and Popescu [1] show a better accuracy at certain resampling factors on anti-aliased images than on aliased images. Nevertheless, our proposed method still outperforms the compared methods at all downsampling factors.

Kernel stretching as another classic anti-aliasing filter is achieved by convolving the image with an interpolation kernel whose spatial support is stretched [55]. Let φ be the interpolation filter defined in the spatial domain. To downsample an image with resampling factor $r \in (0, 1)$, the interpolation filter is transformed by stretching its support by a factor $1/r$:

$$\varphi_r(x) = r \cdot \varphi(rx) \tag{24}$$

before convolving it with the original image. This is equivalent to reducing the bandwidth by a factor $1/r > 1$ in the frequency domain, therefore attenuating the frequencies above the band limit. We applied our comparison on images resampled with the bicubic interpolator and saved in PNG. Its results are shown in Tab. VI. As can be seen, the resampling detection is much harder. All methods perform globally worse than on images filtered by Gaussian blur. Interestingly though, Popescu and Farid's [1] method performs better than our method for the specific resampling factors 0.65 and 0.75, but worse for all other resampling factors. One possible explanation is that the resampling factors $0.65 \approx 2/3$ and $0.75 = 3/4$ result in a third or a fourth of non-interpolated pixels along each axis in the resampled images, and that the Popescu and Farid's method, involving a classification between interpolated and non-interpolated pixels, is advantageous in detecting resampled images containing a large amount of non-interpolated pixels. When most pixels are interpolated for the other resampling factors, the performance of this method drops significantly while our method still works. Our method works well at higher resampling factors and globally outperforms the compared methods on resampled images with anti-aliasing by kernel stretching.

B. Comparison with supervised methods

Next, we compare our unsupervised method with 10 recent supervised neural-network-based methods [56–65]. We rigorously followed the experimental protocol reported in the work of Singh *et al.* [65] and reproduced our results on the same test sets created from BOSSBase [51] and Dresden [52] datasets: 5332 test images in 256×256 were created from 1333 images of BOSSBase, and 5332 test images in 256×256 were created from 214 images of Dresden. It is worth noting that all these compared supervised methods may suffer from resampling-setting bias between training and test, whereas our unsupervised method does not rely on any specific resampling setting of training set. Therefore, evaluations were performed with two protocols: 1) standard setting where the training and test sets both used same factors in $\{1.2, 1.4, 1.6, 1.8, 2\}$ for resampling; 2) cross-evaluation setting where the training set used factors in $\{1.2, 1.4, 1.6, 1.8, 2\}$ while the test set used different factors in $\{1.3, 1.5, 1.7, 1.9, 2.1\}$. In both settings, bilinear interpolation was used, and all the images were saved in uncompressed PNG format. Since the original Dresden images inherently contain demosaicing traces that resemble a resampling factor of 2 and induce spectral correlations at half the image side, we deliberately suppressed these correlations when evaluating on the Dresden dataset to avoid false positive detection. Bidirectional cross validation was employed for our method, and threshold ϵ was set for 1% of FPR on non-resampled PNG images of RAISE dataset-as was chosen in Tab. IV-to report our classification performance.

The comparison results under both the standard and cross-evaluation settings are reported in Table VII, with the results of the compared methods taken from Tables VII and X in [65]. Under the standard setting, our method outperforms all compared approaches on the BOSSBase dataset. On the Dresden dataset, our method yields the highest accuracy on original images but does not reach the same detection performance as the supervised methods on resampled images. This behavior arises because suppressing correlations at half the image side length also attenuates the positive correlation traces associated with the resampling factor of 2. Nevertheless, our method still achieves near-perfect detection for other scaling factors and successfully detects a subset of images resampled at factor 2 by exploiting the weak secondary correlations introduced by demosaicing after resampling. Under the cross-evaluation setting, our proposed method consistently outperforms all other approaches on both the BOSSBase and Dresden datasets. This advantage stems from the fact that our method is inherently unbiased with respect to resampling settings, whereas supervised approaches exhibit a strong dependency on the specific resampling settings used during training.

C. Comparison on resampled images in the wild

We also performed comparison experiments on images resampled by unknown pipelines in the wild. For this purpose, non-resampled images were taken from the RAISE dataset [50], and resampled images were taken from the FODB dataset [53] which contains JPEG-compressed images captured by 27 different cameras and subsequently post-processed by uncontrolled pipelines of five different social media platforms. Although the exact processing pipelines are opaque to us, downsampling is necessarily involved because post-processed images are smaller than their original counterparts. The resampling factors are between 0.21 and 0.90. To alleviate the bias due to JPEG compression, the non-resampled images from RAISE were compressed in JPEG with quantization matrices sampled from the whole FODB dataset. A TV denoiser was used for residual extraction, and bidirectional cross-validation was applied, assuming the downsampling process behind maintained the original aspect ratio. We compared our method with six detection methods [1, 22, 24, 25, 43, 58] and reported recall values with a threshold set for a false positive rate (FPR) of 1% for processing pipelines of different platforms.

The detection performances are reported in Table VIII. In general, all methods exhibit degraded performance under

	Method	Yu [56]	Chen [57]	Bayar [58]	Yang [59]	Singh [60]	Rana [61]	Zhang [62]	Kadha [63]	Niu [64]	Singh [65]	**Ours**
BOSSBase	original	30.6	23.7	54.9	79.8	79.2	87.9	80.4	78.5	80.8	93.5	**99.0**
	resampled	85.6	62.0	90.7	97.5	97.5	97.8	98.2	98.6	98.6	99.4	**99.9**
Dresden	original	33.4	39.1	23.5	35.5	54.6	92.2	89.7	42.6	89.5	94.8	**98.8**
	resampled	87.8	82.2	84.3	98.3	96.8	99.8	99.7	94.8	99.0	**99.9**	84.3
BOSSBase cross-eval.	original	39.2	32.9	51.5	73.6	74.3	81.6	84.5	52.1	82.7	89.1	**99.0**
	resampled	94.8	89.2	96.9	97.1	98.4	99.3	99.9	95.9	96.5	99.8	**100.0**
Dresden cross-eval.	original	29.2	45.5	33.1	32.5	32.3	57.4	77.2	35.9	76.0	97.3	**98.8**
	resampled	31.3	64.1	81.6	82.4	85.1	99.7	99.7	85.7	96.3	96.7	**99.9**

TABLE VII: Detection accuracy (%) on BOSSBase [51] and Dresden [52] datasets. "Cross-eval." denotes a mismatch between training and testing resampling factors for the compared supervised methods, which does not affect our training-free method.

challenging real-world conditions. This is primarily because the aggressive downsampling ratios in real-world pipelines erase most resampling traces, while their heavy JPEG compressions further obscure the remaining artifacts. Nevertheless, our method still demonstrates a certain level of robustness when detecting resampled images from Instagram, Telegram, and WhatsApp, and it achieves the best average performance compared to competing approaches. It is worth noting that the relatively good performance of Popescu's method on Facebook images may be coincidental, since the most similar synthetic map selected by this method does not correspond to the actual resampling rate. By contrast, our method consistently identifies spectral correlations at distances that align with the true resampling rate.

Platform	Facebook	Instagram	Telegram	Twitter	Whatsapp	avg.
Qiao [43]	0.9	0.5	0.2	0.1	0.7	0.5
Mahdian [25]	0.0	0.0	0.0	0.0	0.0	0.0
Gallagher [22]	0.0	0.0	0.0	0.0	0.0	0.0
Birajdar [24]	0.0	0.0	0.0	0.0	0.0	0.0
Popescu [1]	**8.6**	5.3	0.3	**1.3**	1.6	3.4
Bayar [58]	0.5	0.4	0.4	0.4	0.6	0.5
Ours	0.7	**6.7**	**17.8**	0.9	**18.0**	**8.8**

TABLE VIII: Detection accuracy (in %) at 1% FPR of resampling detection methods on images post-processed by different social media platforms. All images were JPEG-compressed. The best results are highlighted in bold.

D. Robustness to different interpolating filters

In this section, we evaluate the proposed *a contrario* detection method on images resampled with different interpolators, including the nearest-neighbor, bilinear, bicubic and Lanczos interpolators. The downsampled images were aliased images. All the images tested were saved in PNG. With a threshold set for FPR at 1%, the detection accuracies at different resampling factors for each interpolator are shown in Fig. 9. We can see that image resampling with the bilinear interpolator is detectable with an accuracy of almost 100%; the performance for the bicubic interpolator is very similar. The nearest-neighbor resampling is easily detected, except for downsampling at factors close to 1. The closer the resampling factor to 1, the more difficult it is to detect. This is because at a resampling factor close to 1 the nearest-neighbor resampling only affects very few pixels by replacing them with their adjacent pixels, and most of

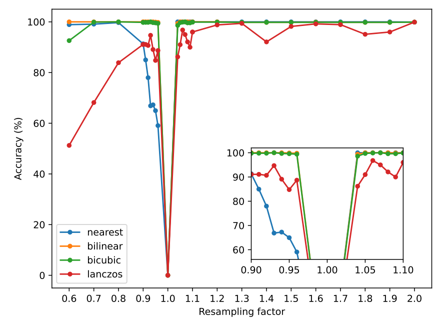


Fig. 9: Detecting accuracies on images resampled with different interpolators at different resampling factors, with an 1% FPR threshold.

the pixels remain unchanged. As for the Lanczos interpolator, which is designed to approximate the sinc filter and attenuates a large amount of aliased frequencies, image resampling is still detectable with lower accuracies. Its detectability increases when the resampling factor increases, so that more than 90% of the upsampled images with the Lanczos interpolator can be detected.

E. Ablation study of residual extraction filters

We carried out an ablation study on the influence of different residual extraction filters on the performance of the detection. Five different residual extraction filters were compared: no residual extraction, the phase-only transform (PHOT) [66], a DCT denoiser [67], the total variation (TV) denoiser [47], and the rank transform [45]. The comparison was conducted on images resampled with the bicubic interpolating filter, then saved respectively in PNG and in JPEG at quality factors 95 and 90. No cross-validation under the proportional resampling assumption was applied.

The detection accuracies with thresholding at 1% FPR are shown in Tab. IX. We can see that all the tested residual extraction filters help boost the detection performance by suppressing the undesirable image contents. Our detector combined with the rank transform performs best on uncompressed images, as the rank transform cancels out the high-level structures of an image and mainly preserves the low-level resampling traces within small neighborhoods. However, the performance on JPEG images using the rank transform is not as good as using a TV denoiser, because the rank transform, which converts a pixel value to its rank in a small neighborhood, is severely perturbed by JPEG blocking artifacts. As there are sudden

	Resampling factor	0.6	0.7	0.8	0.9	1.1	1.2	1.3	1.4	1.5	1.6	average
Uncompressed	none	14.5	50.3	92.0	85.9	98.4	99.9	99.6	98.5	99.3	99.6	83.8
	phase only transform [66]	15.6	48.0	93.6	92.2	99.2	**100.0**	**99.9**	99.1	99.4	99.7	84.7
	DCT denoiser [67]	53.9	83.2	97.2	98.1	99.5	**100.0**	**99.9**	**99.9**	**99.8**	**99.8**	93.1
	TV denoiser [47]	62.6	92.9	98.3	98.7	99.4	99.9	99.4	99.2	99.3	99.7	94.9
	rank transform [45]	**93.1**	**99.9**	**100.0**	**99.8**	**99.7**	99.9	99.8	99.8	**99.8**	**99.8**	**99.2**
JPEG Q=95	none	12.4	41.1	-	74.4	93.2	97.3	90.1	89.7	95.8	-	74.2
	phase only transform [66]	33.8	71.3	-	91.2	95.6	98.0	95.4	93.6	97.7	-	84.6
	DCT denoiser [67]	13.7	39.9	-	82.7	96.3	98.3	94.8	94.1	97.8	-	77.2
	TV denoiser [47]	**48.0**	**78.1**	-	**93.0**	**98.0**	**98.9**	**97.5**	**98.1**	**99.6**	-	**88.9**
	rank transform [45]	47.9	73.6	-	85.0	95.3	97.3	92.7	97.1	99.4	-	86.0
JPEG Q=90	none	6.3	21.5	-	42.3	73.4	86.4	67.9	64.9	79.3	-	55.2
	phase only transform [66]	8.7	24.8	-	58.9	86.1	**93.1**	80.6	80.8	90.8	-	65.5
	DCT denoiser [67]	21.7	45.7	-	**74.9**	87.2	91.0	82.0	85.2	92.3	-	72.5
	TV denoiser [47]	26.6	41.3	-	66.8	**88.2**	**93.1**	**83.9**	**86.9**	**96.5**	-	**72.9**
	rank transform [45]	**30.2**	**46.3**	-	60.5	82.0	85.3	72.5	80.1	91.6	-	68.6

TABLE IX: Detecting accuracy (%) at 1% false positive rate with different residual extraction filters and at resampling factors ranging from 0.6 to 1.6. Resampled images are respectively saved in PNG format and JPEG format at qualities 95 and 90.

changes between the pixels near the JPEG grid, the ranks of these pixels do not truly preserve the local structure and do not contain the useful resampling detection traces. On JPEG-compressed images, the detection applied on the residuals obtained by the TV denoiser achieves the best performance.

Furthermore, we studied the influence of the threshold inherent to these residual extraction filters. The threshold ϵ related to Eq. 21 defines an upper bound of the expected number of false positive distances of correlation in a random spectrum. Whenever a correlation distance is detected in a spectrum, the image is deemed resampled. Therefore, the order of magnitude of the threshold ϵ should be similar to that of the observed false positive rate if the null hypothesis is completely valid. In practice, the null hypothesis is not valid because local spectral correlations can be caused by the contents of the image, and a more conservative value of ϵ is required to have a false positive rate of 1%. Nevertheless, as the required ϵ for a false positive rate of 1% increases, the null hypothesis becomes more valid and the false positive local correlation in a normal spectrum is less significant. The thresholds set at a false positive rate of 1% for these residual extraction filters and for different post-compressions are summarized in Tab. X. One observes that the residual extraction with the phase-only transform, TV denoising or rank transform increases the threshold value, while DCT denoising decreases the threshold value. This means that the phase-only transform, TV denoising and the rank transform are effective in suppressing false positive local correlations caused by image contents, while DCT denoising augments the false positive local correlations. It is also seen in Tab. IX that the detection performance is actually improved when using DCT denoising, indicating that DCT denoising also augments the true positive local correlations due to image resampling. Finally, among all the tested residual extraction filters, the thresholds for the rank transform are the highest, meaning that it is the most effective filter for suppressing false positive local correlations of image contents.

F. Impact of JPEG image compression

Another evaluation was performed on resampled images with different JPEG compression levels to assess the influence of compression on the performance. The images were resampled at factors from 0.6 to 1.6 with the bicubic interpolator and without anti-aliasing, then saved in PNG or in JPEG at different quality factors Q. The detection accuracies at FPR of 1% including all the resampling factors are shown in Fig. 10. Both the results with and without cross validation are visualized. We can see that resampling is easily detected at compression quality factors above 90, and becomes difficult to detect for quality factors below 80. The improvement by cross-validation is small when the resampling detection is too easy or too difficult along each single axis, but when the resampling traces are moderately difficult for Q=80 or Q=70 the cross-validation is helpful to boost the performance.

Threshold ϵ at FPR=1%	PNG	JPEG Q=95	JPEG Q=90
none	1.2×10^{-15}	1.6×10^{-14}	8.5×10^{-17}
phase only transform	2.3×10^{-12}	2.6×10^{-11}	2.6×10^{-11}
DCT denoising	4.0×10^{-19}	5.8×10^{-18}	8.5×10^{-17}
TV denoising	2.2×10^{-7}	1.8×10^{-6}	1.8×10^{-6}
rank transform	1.4×10^{-5}	1.4×10^{-5}	9.7×10^{-5}

TABLE X: NFA thresholds set at 1% false positive rate for different residual extraction filters and different compression rates.

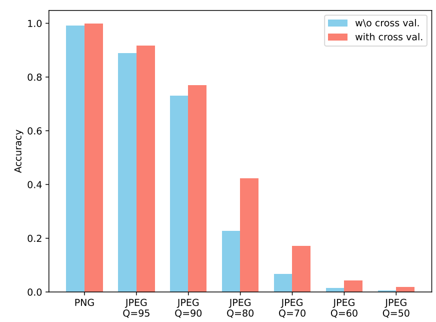


Fig. 10: Accuracy of detecting resampled images saved in PNG or compressed by JPEG at different quality factors Q. Thresholds are set for 1% false positive rate. The performances of the detection solely along the horizontal axis without cross-validation and that with cross-validation by assuming proportional resampling are both shown.

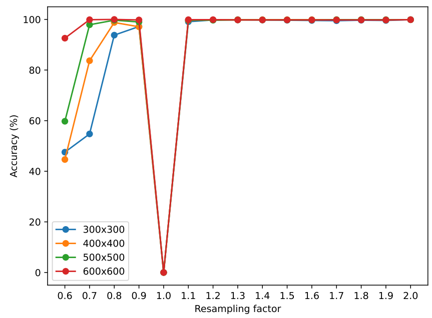


Fig. 11: Accuracy of classifying resampled images of different sizes with threshold set for 1% false positive rate for different resampling factors.

G. Impact of image size

Finally, an evaluation was performed on resampled images of different sizes to assess the influence of image size on detection performance. The images were resampled at factors from 0.6 to 2.0 with the bicubic interpolator and without anti-aliasing, then saved in PNG. The detection accuracies at 1% FPR for different image sizes are shown in Fig. 11. We can see that the detection accuracy increases as the image size increases. The detection of downsampling is degraded on small images, but the detection of upsampling is achievable at an accuracy of almost 100% even on the smallest size of 300×300.

H. Detection on AI-resized images

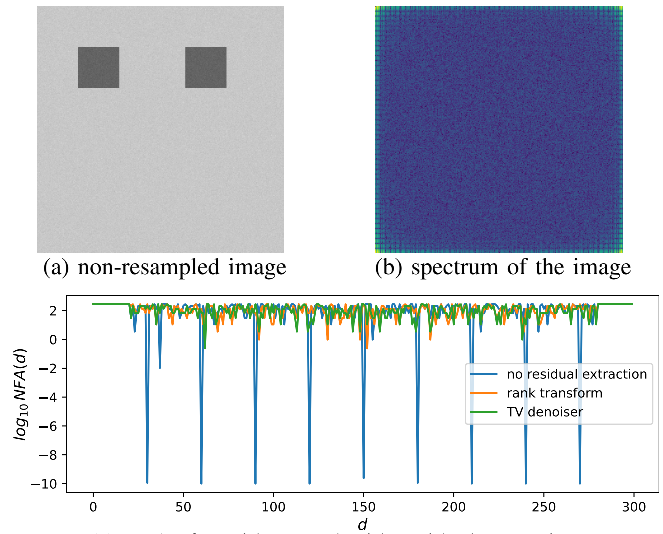
Recent advances in image super-resolution (SR) using deep learning have led to the development of AI-resizing methods that can enhance image resolution with impressive quality. We further investigated the effectiveness of our method on images resized by six different AI-based super-resolution methods: HYPIR [68], InvSR [69], Real-ESRGAN [70], ResShift [71], SwinIR [72], and Swin2SR [73]. We constructed a dataset containing 1,000 original images in 1024×1024 without demosaicing traces from the RAISE dataset, and their corresponding AI-resized images for each SR model. The AI-resized images were generated by applying each SR model to downsampled versions of the original images at scaling factors of 2 and 4. All images were saved in PNG format. We evaluated our detection method using both the rank transform and the TV denoiser for residual extraction. As before, we fixed a threshold at 1% false positive rate on the original images and computed the accuracies on the AI-resized images with the results shown in Tab. XI. It is observed that our method achieves near-perfect performance in distinguishing AI-resized images from non-resized images across all tested SR methods, demonstrating its effectiveness beyond traditional interpolation methods. This is because linear interpolation is a common and necessary step in AI-resizing pipelines, which inevitably introduces spectral correlations that are detectable by our method.

I. Qualitative results

In Section III-A, we discussed the necessity of applying a residual extraction filter prior to detection to suppress false

SR method		HYPIR [68]	InvSR [69]	Real-ESRGAN [70]	ResShift [71]	SwinIR [72]	Swin2SR [73]
$\times 2$	TV	100.0	98.4	100.0	100.0	100.0	100.0
	RT	100.0	99.8	100.0	100.0	100.0	100.0
$\times 4$	TV	100.0	100.0	100.0	100.0	100.0	100.0
	RT	100.0	100.0	100.0	100.0	100.0	100.0

TABLE XI: Detecting accuracy (%) at 1% false positive rate on AI-resized images by different SR methods at scaling factors 2 and 4. Images are saved in PNG format. Both the TV denoiser (TV) and the rank transform (RT) as residual extraction filters are tested.



(a) non-resampled image (b) spectrum of the image

(c) NFAs for without and with residual extraction

Fig. 12: Analysis of a simulated image containing two identical squares with additive Gaussian noise. Replicated scene content leads to periodic patterns in the spectrum and false positive correlations (low NFAs) when no residual extraction is applied, which are effectively eliminated in the residual of the rank transform or the TV denoiser.

detections caused by image contents. Here we show a simulated example with two squares of the same size and additive Gaussian noise in Fig. 12. Replicated patterns are visually present in its spectrum and cause strong local correlations at specific distances. These spurious correlations are mistakenly detected as resampling traces by our detector when no residual extraction is applied, and are effectively suppressed in the image residual extracted by the rank transform or the TV denoiser.

Qualitative analyses in Figs. 13–15 further illustrate the efficacy of various residual extraction filters. In Fig. 13 where textures are smooth, resampling traces are easily detected in both the original image and its residual from any filter. However, when an image contain diverse and highly contrasted content (Fig. 14), inherent image correlations may mask those introduced by resampling; in such cases, subtle artifacts only become detectable after enhancement via DCT denoising, TV denoising, or rank transform. Furthermore, Fig. 15 highlights the role of cross-validation in non-resampled images. While replicated structures like pillars can trigger false positives by creating strong spectral correlations at certain distances, these

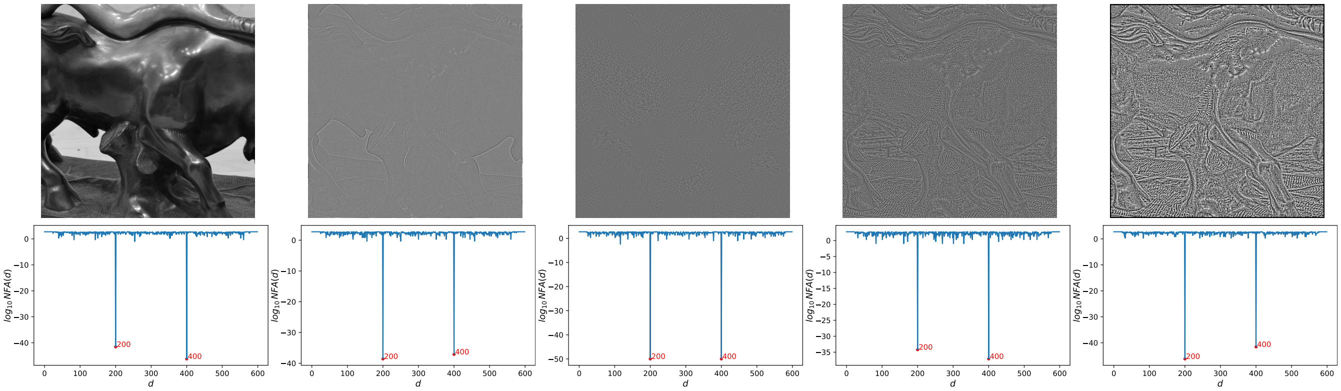


Fig. 13: Detection results with different residual extraction filters on an image resampled from 1000×1000 to 600×600. Top: images after residual extraction; bottom: the NFAs of different distances d. The expected positive correlation distances d at 200 and 400 are detected either without or with any residual extraction filter.

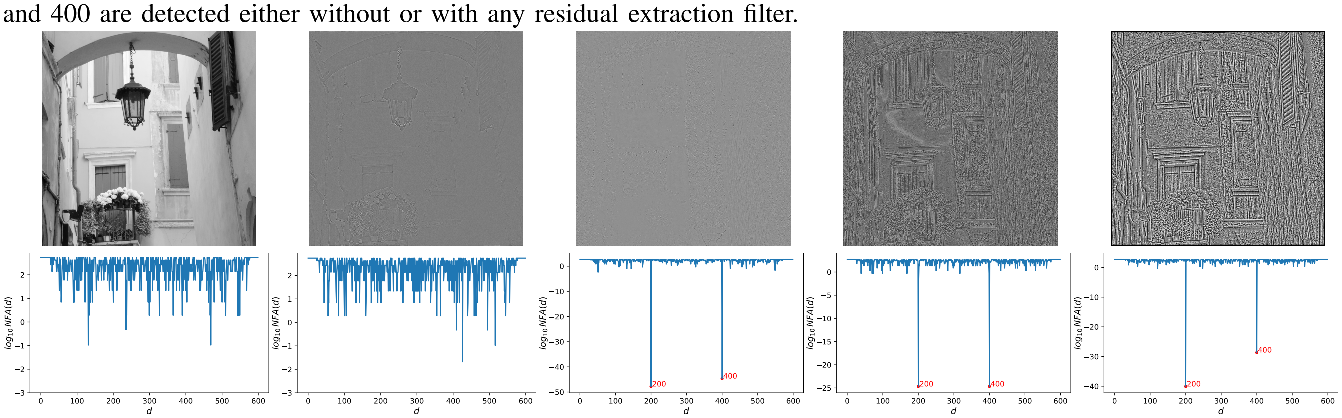


Fig. 14: Detection results with different residual extraction filters on an image resampled from 1000×1000 to 600×600. Top: images after residual extraction; bottom: the NFAs of different distances d. The expected positive correlation distances d at 200 and 400 are detected only with DCT denoiser, TV denoiser, and rank transform.

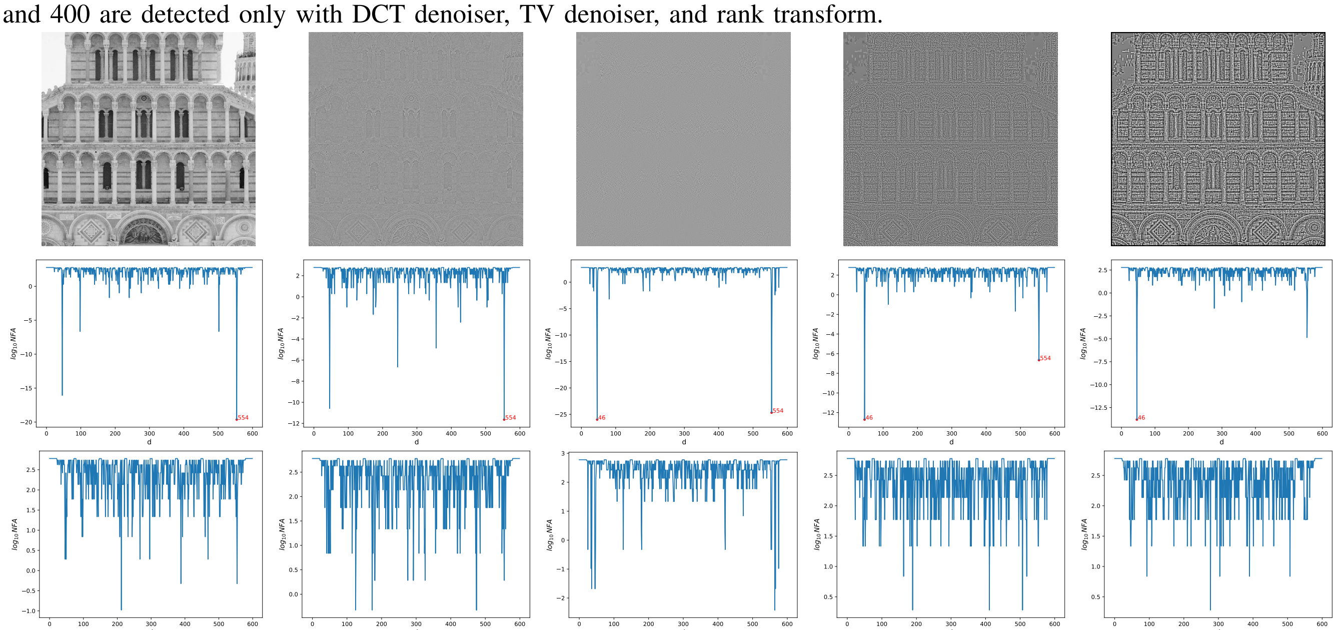


Fig. 15: Detection results with different residual extraction filters on a *non-resampled* image. Top: images after residual extraction; middle: the NFAs of different distances d only along the horizontal axis; bottom: the NFAs of different distances d with cross validation by assuming proportional resampling. Directional spectral correlations caused by replicated pillars are filtered out by enforcing symmetry between horizontal and vertical resampling traces.

artifacts typically appear along a single axis (e.g., horizontal). By imposing the proportional resampling assumption through cross-validation, these directional false positives are successfully suppressed, as they lack the required symmetry across both dimensions.

V. Discussion and Conclusion

In this paper, we conducted a theoretical study of the spectrum folding phenomenon for both one-dimensional signal resampling and two-dimensional image resampling. We demonstrated that local spectral correlations induced by resampling could be effectively exploited for detection purposes. To enhance the detectability of these correlations and suppress false positives caused by image content, we proposed a residual extraction step utilizing non-linear filters. We then developed an *a contrario* method to detect anomalous correlations between small patch pairs within the spectrum. The method was further adapted for JPEG images by mitigating false positives arising from JPEG compression artifacts, and for proportionally resampled images along both horizontal and vertical directions through a cross-validation strategy. Experimental results showed that our method outperformed existing techniques by a large margin across diverse scenarios, ranging from standard PNG/JPEG formats and aliasing levels to AI-resized and real-world social media images. Finally, a comprehensive study was carried out to analyze how various factors—including interpolation kernels, residual extraction filters, JPEG quality factors, and image size—influenced the overall detection performance.

Nevertheless, the proposed method has some limitations in its current form. First, resampling at specific factors in the form of $\{8/z \mid z \in \mathbb{N}^*\}$ can be systematically masked by posterior JPEG compression, making them blind to our proposed method and to other unsupervised methods [1, 22, 24, 25] as well. Second, the presented spectrum folding argument does not allow us to determine the original size M of a resampled image from the resulting correlation offset d and the current size N due to the ambiguity of M retrieved from $d = \pm M \bmod N$. More work is needed to find an effective disambiguation method. In addition, the presented spectrum folding argument is currently limited to image rescaling where the image is resampled along each axis independently at different rates. In practice, image resampling is also used for image skewing and rotation, which has been shown to be detectable in previous works [1, 25]. Our theory needs to be extended to general affine transformations to address these cases. Last but not least, despite its effectiveness in detecting AI-resized in uncompressed formats, it currently struggles with JPEG-compressed versions, as both authentic JPEG artifacts and AI-resizing introduce spectral correlations at identical distances.

Beyond these challenges, the current frequency-domain analysis relies on the full image and is unable to detect local resampled zones. One way of detecting local resampled zones with our proposed method is to process the algorithm on small patches and assemble the local detection results to locate resampled zones. A last inconvenience is the high computational cost of the proposed *a contrario* algorithm, whose time complexity is $\mathcal{O}(L^3 \frac{l}{\log(L)})$ for an image of size $L \times L$ and patch size $l \times l$. This can be potentially improved by combining faster methods such as those in [22, 24, 25] with thresholding at high recalls to provide candidate correlation distances, then restricting the application of the *a contrario* algorithm to sparse candidate distances.

Acknowledgments

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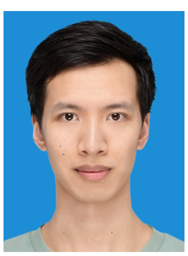

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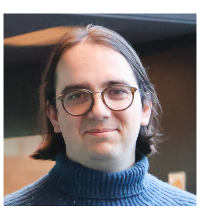

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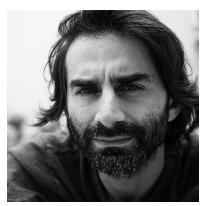

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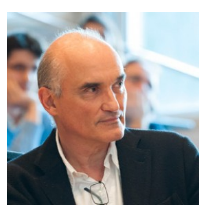

Jean-Michel Morel was awarded in 1980 a PhD by University Marie et Pierre Curie. He has been an assistant professor at Université Paris-Dauphine, a full professor at Ecole Normale Supérieure Paris-Saclay and is currently chair professor in imaging science at Lingnan University, Hong Kong. His main interests are the mathematical theory and the algorithms of image analysis and image processing. Recent awards: INRIA - French Academy of Science Prize 2013, CNRS Innovation Medal 2015, IEEE Longuet-Higgins Prize 2015.

Appendix A
Theorem: Modulated spectrum folding of aligned resampling in 1D

In the following, we adopt the notation conventions established in Sec. II.

Consider a 1D signal $\{u_k\} \in \mathbb{R}^M$ of M samples and its resampled version $\{v_k\} \in \mathbb{R}^N$ of N samples. Here $M, N \in \mathbb{N}^*$ can be any positive integers. An usual resampling method proceeds in two steps: first the samples $\{u_k\}$ are interpolated to a continuous function f using an interpolating filter φ, then f is discretized to the signal $\{v_k\}$ of N samples.

$$\{u_k\} \in \mathbb{R}^M \xrightarrow[\varphi]{\text{interpolation}} f \xrightarrow{\text{discretization}} \{v_k\} \in \mathbb{R}^N$$

The interpolation from $\{u_k\}$ to a continuous function $f : \mathbb{R} \to \mathbb{R}$ is realized by convolving $\{u_k\}$ with an interpolating filter φ:

$$f(x) = \sum_{k=0}^{M-1} u_k \cdot \varphi(x-k), \qquad x \in \mathbb{R}. \tag{S1}$$

Then the resampling to $\{v_k\}$ is achieved by evenly discretizing f on $[0, M]$ to N samples:

$$v_k = f\left(k\frac{M}{N}\right), \qquad k = 0, \ldots, N-1. \tag{S2}$$

More specifically, such resampling is an aligned resampling because the intervals for the original sequence and resampled sequence are the same. An illustrative example is shown in Fig. S1.

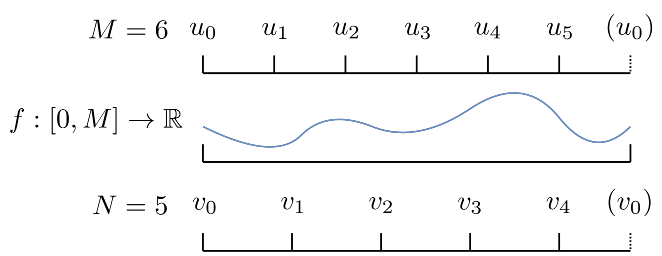

Fig. S1: Aligned resampling: the intervals for the original sequence and the resampled sequence are the same.

Then we have the following theorem for the 1D aligned resampling:

Theorem (Modulated Spectrum Folding in 1D)**.** *Suppose the interpolating filter φ is M-periodic. Denote the DFTs of $\{u_k\}$ and $\{v_k\}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Then,*

$$\tilde{v}_n = M \cdot \sum_{z \in \mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n}, \tag{S3}$$

where $\{c_p(\varphi)\}_{p\in\mathbb{Z}}$ denotes the Fourier coefficients of φ.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be the interpolation of $\{u_k\}$ obtained by convolving $\{u_k\}$ with the interpolating filter φ following Eq. S1. Let $\{c_p(f)\}_{p\in\mathbb{Z}}$ be the Fourier coefficients of f, i.e.:

$$f(x) = \sum_{p\in\mathbb{Z}} c_p(f)\omega_M^{xp}. \tag{S4}$$

The first step is to establish the relation between $\{\tilde{u}_n\}$, $\{c_p(\varphi)\}$ and $\{c_p(f)\}$ during the interpolation process and to show that $c_p(f) = M c_p(\phi)\tilde{u}_p$. By the definition of f as the convolution of $\{u_k\}$ with φ and replacing those with their Fourier expressions, we have:

$$\begin{aligned} f(x) &= \sum_{k=0}^{M-1} u_k \cdot \varphi(x-k) \\ &= \sum_{k=0}^{M-1} \left(\sum_{m=0}^{M-1} \tilde{u}_m \cdot \omega_M^{km}\right) \cdot \left(\sum_{p\in\mathbb{Z}} c_p(\varphi) \cdot \omega_M^{(x-k)p}\right). \end{aligned} \tag{S5}$$

Rearranging the terms, we can write:

$$\begin{aligned} f(x) &= \sum_{k=0}^{M-1}\sum_{m=0}^{M-1}\sum_{p\in\mathbb{Z}} \tilde{u}_m \cdot \omega_M^{km} \cdot c_p(\varphi) \cdot \omega_M^{(x-k)p} \\ &= \sum_{p\in\mathbb{Z}} \omega_M^{xp} c_p(\varphi) \sum_{m=0}^{M-1} \tilde{u}_m \sum_{k=0}^{M-1} \omega_M^{k(m-p)}. \end{aligned} \tag{S6}$$

Note that:

$$\sum_{k=0}^{M-1} \omega_M^{k(m-p)} = \begin{cases} M & \text{if } m = p (mod M) \\ 0 & \text{otherwise} \end{cases},$$

which yields:

$$f(x) = \sum_{p\in\mathbb{Z}} \omega_M^{xp} c_p(\varphi) M \tilde{u}_{p \bmod M} = \sum_{p\in\mathbb{Z}} \omega_M^{xp} c_p(\varphi) M \tilde{u}_p. \tag{S7}$$

By identification from Eq. S7 and the definition of Fourier series $f(x) = \sum_{p\in\mathbb{Z}} \omega_M^{xp} \cdot c_p(f)$, we get

$$c_p(f) = M c_p(\varphi)\tilde{u}_p. \tag{S8}$$

The second step is to relate $\{\tilde{v}_n\}$ and $\{c_p(f)\}$ during the discretization process and to show that $\tilde{v}_n = \sum_{z\in\mathbb{Z}} c_{zN+n}(f)$. By the definition of $\{v_k\}$ in Eq. S2 and the Fourier series of f in Eq. S17, we have

$$v_k = f\left(k\frac{M}{N}\right) \text{ for all } k = 0, \ldots, N-1, \tag{S9}$$

which can be written as:

$$v_k = \sum_{p\in\mathbb{Z}} c_p(f)\omega_M^{k\frac{M}{N}p} = \sum_{p\in\mathbb{Z}} c_p(f)\omega_N^{kp}. \tag{S10}$$

Plugging this expression in the DFT definition of $\{\tilde{v}_n\}$, we get:

$$\begin{aligned} \tilde{v}_n &= \frac{1}{N}\sum_{k=0}^{N-1} v_k \omega_N^{-nk} \\ &= \frac{1}{N}\sum_{k=0}^{N-1}\left(\sum_{p\in\mathbb{Z}} c_p(f)\omega_N^{kp}\right)\omega_N^{-nk}. \end{aligned} \tag{S11}$$

Rearranging the terms, we get:

$$\tilde{v}_n = \sum_{p\in\mathbb{Z}} c_p(f)\left(\frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{k(p-n)}\right). \tag{S12}$$

Note again that:

$$\frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{k(p-n)}=\begin{cases}1 & \text{if } p=n(modN)\\ 0 & \text{otherwise}\end{cases},$$

which yields that the surviving terms in Eq. S12 have indices $p\in\{zN+n|z\in\mathbb{Z}\}$. This allows us to condensate the expression of $\tilde{v}_n$ as

$$\tilde{v}_n=\sum_{z\in\mathbb{Z}}c_{zN+n}(f). \tag{S13}$$

Finally, combining Eqs. S8 and S13, we obtain:

$$\tilde{v}_n=M\cdot\sum_{z\in\mathbb{Z}}c_{zN+n}(\varphi)\cdot\tilde{u}_{zN+n}, \tag{S14}$$

□

APPENDIX B
THEOREM: MODULATED SPECTRUM FOLDING OF ALIGNED RESAMPLING IN MULTIDIMENSIONS

Consider an original d-dimensional signal of size $M_1\times\ldots\times M_d$: $\mathbf{u}=\{(u_{k_1,\ldots,k_d})\}\in\mathbb{R}^{M_1\times\ldots\times M_d}$. $\mathbf{u}$ is resampled to another d-dimensional signal $\mathbf{v}=\{(v_{k_1,\ldots,k_d})\}\in\mathbb{R}^{N_1\times\ldots\times N_d}$ using an interpolating filter $\varphi:\mathbb{R}^d\to\mathbb{R}^d$.

$$\{u_k\}\in\mathbb{R}^{M_1\times\ldots\times M_d}\xrightarrow[\varphi]{\text{interpolation}}f\xrightarrow{\text{discretization}}\{v_k\}\in\mathbb{R}^{N_1\times\ldots\times N_d}$$

Suppose the interpolating filter φ is $(M_1,\ldots,M_d)$-periodic, *i.e.* for any $1\le i\le d$, $\varphi(\ldots,x_i,\ldots)=\varphi(\ldots,x_i+M_i,\ldots)$. $\mathbf{u}$ is interpolated to function $f:\mathbb{R}^d\to\mathbb{R}^d$ by convolution with φ in d-dimension:

$$f(x_1,\ldots,x_d)=\sum_{k_1=0}^{M_1-1}\ldots\sum_{k_d=0}^{M_d-1}u_{k_1,\ldots,k_d}\,\varphi(x_1-k_1,\ldots,x_d-k_d). \tag{S15}$$

And f is discretized evenly on $[0,M_1]\times\ldots\times[0,M_d]$ to $\mathbf{v}$ by taking $v_{k_1,\ldots,k_d}=f(k_1\frac{M_1}{N_1},\ldots,k_d\frac{M_d}{N_d})$.

Similarly to 1D, such defined resampling is an aligned resampling because the original signal $\mathbf{u}$ and the resampled signal $\mathbf{v}$ share the same interval in each dimension.

Then we have the following theorem for multidimensional aligned resampling:

Theorem (Modulated Spectrum Folding in Multidimensions)**.** *Suppose the interpolating filter φ is $(M_1,\ldots,M_d)$-periodic, i.e. for any $1\le i\le d$, $\varphi(\ldots,x_i,\ldots)=\varphi(\ldots,x_i+M_i,\ldots)$. Denote the DFTs of $\mathbf{u}$ and $\mathbf{v}$ by $\{\tilde{u}_{\mathbf{m}}\}\in\mathbb{C}^{M_1\times\ldots\times M_d}$ with $\mathbf{m}:=(m_1,\ldots,m_d)$ and $\{\tilde{v}_{\mathbf{n}}\}\in\mathbb{C}^{N_1\times\ldots\times N_d}$ with $\mathbf{n}:=(n_1,\ldots,n_d)$, respectively. Denote the Fourier series of φ by $\{c_{\mathbf{p}}(\varphi)\}_{\mathbf{p}\in\mathbb{Z}^d}$ with $\mathbf{p}=(p_1,\ldots,p_d)$. Then,*

$$\tilde{v}_{\mathbf{n}}=\left(\prod_{i=1}^{d}M_i\right)\cdot\sum_{\mathbf{z}\in\mathbb{Z}^d}c_{\mathbf{z}\odot\mathbf{N}+\mathbf{n}}(\varphi)\cdot\tilde{u}_{[(\mathbf{z}\odot\mathbf{N}+\mathbf{n})\bmod\mathbf{M}]} \tag{S16}$$

where $\mathbf{M}:=(M_1,\ldots,M_d)$ and $\mathbf{N}:=(N_1,\ldots,N_d)$. $\bmod$ and $\odot$ is element-wise modulo and multiplication, respectively.

Proof. The proof is similar to the case of 1-dimension. Let $f:\mathbb{R}^d\to\mathbb{R}^d$ be the interpolation of $\{u_k\}$ obtained by convolving $\mathbf{u}$ with the interpolating filter φ following Eq. S15. Let $\{c_p(f)\}_{p\in\mathbb{Z}}$ be the Fourier coefficients of f, i.e.:

$$f(x)=\sum_{p\in\mathbb{Z}}c_p(f)\omega_M^{xp}. \tag{S17}$$

According to the definitions of DFT of $\mathbf{u}$ and $\mathbf{v}$ and the Fourier series of φ and f in multidimensions, we have:

$$u_{k_1,\dots,k_d} = \sum_{m_1=0}^{M_1-1}\Big(\omega_{M_1}^{k_1 m_1}\sum_{m_2=0}^{M_2-1}\Big(\omega_{M_2}^{k_2 m_2}\dots \sum_{m_d=0}^{M_d-1}\omega_{M_d}^{k_d m_d}\cdot \tilde{u}_{m_1,\dots,m_d}\Big)\Big) \tag{S18}$$

$$v_{k_1,\dots,k_d} = \sum_{n_1=0}^{N_1-1}\Big(\omega_{N_1}^{k_1 n_1}\sum_{n_2=0}^{N_2-1}\Big(\omega_{N_2}^{k_2 n_2}\dots \sum_{n_d=0}^{N_d-1}\omega_{N_d}^{k_d n_d}\cdot \tilde{v}_{n_1,\dots,n_d}\Big)\Big) \tag{S19}$$

$$\varphi(x_1,\dots,x_d) = \sum_{p_1\in\mathbb{Z}}\Big(\omega_{M_1}^{x_1 p_1}\sum_{p_1\in\mathbb{Z}}\Big(\omega_{M_2}^{x_2 p_2}\dots \sum_{p_d\in\mathbb{Z}}\omega_{M_d}^{x_d p_d}\cdot c_{p_1,\dots,p_d}(\varphi)\Big)\Big) \tag{S20}$$

$$f(x_1,\dots,x_d) = \sum_{p_1\in\mathbb{Z}}\Big(\omega_{M_1}^{x_1 p_1}\sum_{p_1\in\mathbb{Z}}\Big(\omega_{M_2}^{x_2 p_2}\dots x\sum_{p_d\in\mathbb{Z}}\omega_{M_d}^{x_d p_d}\cdot c_{p_1,\dots,p_d}(f)\Big)\Big) \tag{S21}$$

By setting $\omega_{\mathbf{M}}^{\mathbf{x}} := e^{2\pi i(\mathbf{x}^\top\cdot\frac{1}{\mathbf{M}})}$, $\mathcal{M} := [\![0, M_1-1]\!]\times\dots\times[\![0, M_d-1]\!]$ and $\mathcal{N} := [\![0, N_1-1]\!]\times\dots\times[\![0, N_d-1]\!]$, we simplify the above formulas as:

$$u_{\mathbf{k}} = \sum_{\mathbf{m}\in\mathcal{M}}\tilde{u}_{\mathbf{m}}\cdot\omega_{\mathbf{M}}^{\mathbf{k}\odot\mathbf{m}} \tag{S22}$$

$$v_{\mathbf{k}} = \sum_{\mathbf{n}\in\mathcal{N}}\tilde{v}_{\mathbf{n}}\cdot\omega_{\mathbf{N}}^{\mathbf{k}\odot\mathbf{n}} \tag{S23}$$

$$\varphi(\mathbf{x}) = \sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(\varphi)\cdot\omega_{\mathbf{M}}^{\mathbf{x}\odot\mathbf{p}} \tag{S24}$$

$$f(\mathbf{x}) = \sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(f)\cdot\omega_{\mathbf{M}}^{\mathbf{x}\odot\mathbf{p}} \tag{S25}$$

The first step is to relate the Fourier coefficients of f to the DFT of $\mathbf{u}$ and the Fourier coefficients of φ during the interpolation process. Starting from the definition of f in Eq. S15, we replace the terms $u_{\mathbf{k}}$ and $\varphi(\mathbf{x})$ by their Fourier expressions:

$$\begin{aligned} f(\mathbf{x}) &= \sum_{\mathbf{k}\in\mathcal{M}}u_{\mathbf{k}}\,\varphi(\mathbf{x}-\mathbf{k}) \\ &= \sum_{\mathbf{k}\in\mathcal{M}}\left(\sum_{\mathbf{m}\in\mathcal{M}}\tilde{u}_{\mathbf{m}}\cdot\omega_{\mathbf{M}}^{\mathbf{k}\odot\mathbf{m}}\right)\cdot\left(\sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(\varphi)\cdot\omega_{\mathbf{M}}^{(\mathbf{x}-\mathbf{k})\odot\mathbf{p}}\right) \\ &= \sum_{\mathbf{k}\in\mathcal{M}}\sum_{\mathbf{m}\in\mathcal{M}}\sum_{\mathbf{p}\in\mathbb{Z}^d}\tilde{u}_{\mathbf{m}}\cdot\omega_{\mathbf{M}}^{\mathbf{k}\odot\mathbf{m}}\cdot c_{\mathbf{p}}(\varphi)\cdot\omega_{\mathbf{M}}^{(\mathbf{x}-\mathbf{k})\odot\mathbf{p}} \\ &= \sum_{\mathbf{p}\in\mathbb{Z}^d}\omega_{\mathbf{M}}^{\mathbf{x}\odot\mathbf{p}}\underbrace{\sum_{\mathbf{k}\in\mathcal{M}}\sum_{\mathbf{m}\in\mathcal{M}}\tilde{u}_{\mathbf{m}}\cdot\omega_{\mathbf{M}}^{\mathbf{k}\odot\mathbf{m}}\cdot c_{\mathbf{p}}(\varphi)\cdot\omega_{\mathbf{M}}^{-\mathbf{k}\odot\mathbf{p}}}_{c_{\mathbf{p}}(f)} \end{aligned} \tag{S26}$$

By identification from Eq. S25 we have

$$\begin{aligned} c_{\mathbf{p}}(f) &= \sum_{\mathbf{k}\in\mathcal{M}}\sum_{\mathbf{m}\in\mathcal{M}}\tilde{u}_{\mathbf{m}}\cdot\omega_{\mathbf{M}}^{\mathbf{k}\odot\mathbf{m}}\cdot c_{\mathbf{p}}(\varphi)\cdot\omega_{\mathbf{M}}^{-\mathbf{k}\odot\mathbf{p}} \\ &= c_{\mathbf{p}}(\varphi)\sum_{\mathbf{m}\in\mathcal{M}}\tilde{u}_{\mathbf{m}}\sum_{\mathbf{k}\in\mathcal{M}}\omega_{\mathbf{M}}^{\mathbf{k}\odot(\mathbf{m}-\mathbf{p})} \end{aligned} \tag{S27}$$

As $\mathbf{p}\in\mathbb{Z}^d$ and $\mathbf{m}\in[\![0, M_1-1]\!]\times\dots\times[\![0, M_d-1]\!]$, $\sum_{\mathbf{k}\in\mathcal{M}}\omega_{\mathbf{M}}^{\mathbf{k}\odot(\mathbf{m}-\mathbf{p})} = \left(\prod_{i=1}^d M_i\right)\mathbb{1}_{\{\mathbf{p}\bmod\mathbf{M}=\mathbf{m}\}}$. Indeed,

$$\begin{aligned} \sum_{\mathbf{k}\in\mathcal{M}}\omega_{\mathbf{M}}^{\mathbf{k}\odot(\mathbf{m}-\mathbf{p})} &= \sum_{k_1=0}^{M_1-1}\dots\sum_{k_d=0}^{M_d-1}\omega_{M_1}^{k_1(m_1-p_1)}\dots\omega_{M_d}^{k_d(m_d-p_d)} \\ &= \left(\sum_{k_1=0}^{M_1-1}\omega_{M_1}^{k_1(m_1-p_1)}\right)\dots\left(\sum_{k_d=0}^{M_d-1}\omega_{M_d}^{k_d(m_d-p_d)}\right) \\ &= M_1\mathbb{1}_{\{p_1\bmod M_1=m_1\}}\dots M_d\mathbb{1}_{\{p_d\bmod M_d=m_d\}} \\ &= \left(\prod_{i=1}^d M_i\right)\mathbb{1}_{\{\mathbf{p}\bmod\mathbf{M}=\mathbf{m}\}} \end{aligned} \tag{S28}$$

Eq. S27 is reduced to:

$$\begin{aligned} c_{\mathbf{p}}(f) &= c_{\mathbf{p}}(\varphi)\sum_{\mathbf{m}\in\mathcal{M}}\tilde{u}_{\mathbf{m}}\left(\prod_{i=1}^d M_i\right)\mathbb{1}_{\{\mathbf{p}\bmod\mathbf{M}=\mathbf{m}\}} \\ &= \left(\prod_{i=1}^d M_i\right)c_{\mathbf{p}}(\varphi)\cdot\tilde{u}_{[\mathbf{p}\bmod\mathbf{M}]} \end{aligned} \tag{S29}$$

which gives the relation between $\{\tilde{u}_{\mathbf{m}}\}$, $\{c_{\mathbf{p}}(\varphi)\}$ and $\{c_{\mathbf{p}}(f)\}$.

The second step is to derive the relation between $\{\tilde{v}_{\mathbf{n}}\}$ and $\{c_{\mathbf{p}}(f)\}$. Given the resampling defined as $v_{\mathbf{k}} = f(\mathbf{k}\odot\mathbf{M}\odot\frac{1}{\mathbf{N}})$, we have

$$\begin{aligned} v_{\mathbf{k}} &= f(\mathbf{k}\odot\mathbf{M}\odot\frac{1}{\mathbf{N}}) \\ &= \sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(f)\cdot\omega_{\mathbf{M}}^{\mathbf{k}\odot\mathbf{M}\odot\frac{1}{\mathbf{N}}\odot\mathbf{p}} \\ &= \sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(f)\cdot e^{2\pi i\left(\left(\mathbf{k}\odot\mathbf{M}\odot\frac{1}{\mathbf{N}}\odot\mathbf{p}\right)^\top\cdot\frac{1}{\mathbf{M}}\right)} \\ &= \sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(f)\cdot e^{2\pi i\left((\mathbf{k}\odot\mathbf{p})^\top\cdot\frac{1}{\mathbf{N}}\right)} \\ &= \sum_{\mathbf{p}\in\mathbb{Z}^d}c_{\mathbf{p}}(f)\cdot\omega_{\mathbf{N}}^{\mathbf{k}\odot\mathbf{p}} \\ &\text{(split } \mathbb{Z}^d \text{ into disjoint hyper-cubes of size } N_1\times\dots\times N_d) \\ &= \sum_{\mathbf{z}\in\mathbb{Z}^d}\sum_{\mathbf{n}\in\mathcal{N}}c_{\mathbf{z}\odot\mathbf{N}+\mathbf{n}}(f)\cdot\omega_{\mathbf{N}}^{\mathbf{k}\odot(\mathbf{z}\odot\mathbf{N}+\mathbf{n})} \\ &= \sum_{\mathbf{n}\in\mathcal{N}}\underbrace{\sum_{\mathbf{z}\in\mathbb{Z}^d}c_{\mathbf{z}\odot\mathbf{N}+\mathbf{n}}(f)}_{\tilde{v}_{\mathbf{n}}}\cdot\omega_{\mathbf{N}}^{\mathbf{k}\odot\mathbf{n}} \end{aligned} \tag{S30}$$

By identification from Eqs. S23 and S30, we retrieve the expression of $\tilde{v}_{\mathbf{n}}$:

$$\tilde{v}_{\mathbf{n}} = \sum_{\mathbf{z}\in\mathbb{Z}^d}c_{\mathbf{z}\odot\mathbf{N}+\mathbf{n}}(f) \tag{S31}$$

Finally, plugging the expression of $c_{\mathbf{p}}(f)$ of Eq. S29 into Eq. S31 yields:

$$\tilde{v}_{\mathbf{n}} = \left(\prod_{i=1}^{d} M_i\right) \sum_{\mathbf{z}\in\mathbb{Z}^d} c_{\mathbf{z}\odot\mathbf{N}+\mathbf{n}}(\varphi) \cdot \tilde{u}_{[(\mathbf{z}\odot\mathbf{N}+\mathbf{n}) \bmod \mathbf{M}]}. \tag{S32}$$

□

APPENDIX C
COVARIANCE BETWEEN TWO ARBITRARY COEFFICIENTS IN A SPECTRUM OF RESAMPLED WHITE NOISE

Theorem. *Let the original signal* $\{u_k\} \in \mathbb{R}^M$ *be a sequence of white noise with* $\mathrm{Var}[u_k] = \sigma^2$ *and* $\mathrm{E}[u_k] = 0$*, which is resampled to* $\{v_k\} \in \mathbb{R}^N$ *using an* M*-periodic interpolating filter* $\varphi : \mathbb{R} \mapsto \mathbb{R}$*. Let* $\tilde{v}_m$ *and* $\tilde{v}_n$ *be two arbitrary frequency components in the DFT of the resampled signal,* $m, n \in \mathbb{Z}$*. Denote by* $L = lcm(M,N)$ *and* $gcd(M,M)$ *the least common multiple and the greatest common divisor of* M *and* N*, respectively. We have* $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m]$ *given by:*

- *if* $m-n$ *is not a multiple of* $gcd(M,N)$*,* $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] = 0$*;*
- *if* $m-n$ *is a multiple of* $gcd(M,N)$*, the covariance is given by*

$$\begin{aligned}\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] &= \mathrm{Cov}[\tilde{v}_n, \tilde{v}_{n+kM}] \\ &= M\sigma^2 \sum_{q\in\mathbb{Z}} \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM+qL}(\varphi)}\end{aligned} \tag{S33}$$

where $k = \mathcal{K}(m-n)$ *and* $\mathcal{K}(x)$ *is defined as the solution of*

$$\begin{aligned} \min_{k'} \quad & |k'| \\ \textit{s.t.} \quad & k'M(\bmod N) = x(\bmod N) \\ & k' \in \mathbb{Z} \end{aligned} \tag{S34}$$

Proof. It is easy to show $\mathrm{E}[\tilde{u}_n] = 0$ and $\mathrm{Var}[\tilde{u}_n] = \mathrm{E}[\tilde{u}_n\overline{\tilde{u}_n}] = \frac{\sigma^2}{M}$ for any $n \in \mathbb{Z}$. According to Theorem 1 we have

$$\begin{aligned}\tilde{v}_n &= M \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n} \\ \tilde{v}_m &= M \sum_{z\in\mathbb{Z}} c_{zN+m}(\varphi) \cdot \tilde{u}_{zN+m}\end{aligned} \tag{S35}$$

Naturally, $\mathrm{E}[\tilde{v}_m] = \mathrm{E}[\tilde{v}_n] = 0$. Their covariance is

$$\begin{aligned}&\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] \\ &= M^2\mathrm{E}[(\sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \tilde{u}_{zN+n})\overline{(\sum_{z'\in\mathbb{Z}} c_{z'N+m}(\varphi) \cdot \tilde{u}_{z'N+m})}] \\ &= M^2 \sum_{z\in\mathbb{Z}} \sum_{z'\in\mathbb{Z}} c_{zN+n}(\varphi)\overline{c_{z'N+m}(\varphi)} \cdot \mathrm{E}[\tilde{u}_{zN+n}\overline{\tilde{u}_{z'N+m}}].\end{aligned} \tag{S36}$$

As the DFT $\{\tilde{u}_n\}$ is M-periodic, $E[\tilde{u}_{zN+n}\overline{\tilde{u}_{z'N+m}}] = \frac{\sigma^2}{M}$ if and only if $\exists k' \in \mathbb{Z}$ such that

$$\begin{aligned}& zN + n + k'M = z'N + m \\ \Leftrightarrow\ & N(z - z') + Mk' = m - n,\end{aligned} \tag{S37}$$

Eq. S37 is actually a linear Diophantine equation in the form of $Nx + My = m - n$ where x and y are two integer unknowns. Therefore, if $m - n$ is not a multiple of $gcd(M,N)$, Eq. S37 has no solution and $\mathrm{E}[\tilde{u}_{zN+n}\overline{\tilde{u}_{z'N+m}}] = 0$ for all the $z, z' \in \mathbb{Z}$, which yields $\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] = 0$.

If $m-n$ is a multiple of $gcd(M,N)$, then the possible k' that make Eq. S37 hold satisfy $m - n(\bmod N) = k'M(\bmod N)$. Let $\mathbf{K}$ be the set of the possible values of k' and let $k = \arg\min_{k'\in\mathbf{K}} |k'|$, then the possible $z - z'$ that make Eq. S37 hold are $z - z' \in \{\frac{m-n-kM}{N} + q\frac{L}{N}, q \in \mathbb{Z}\}$. Finally, Eq. S36 is rearranged to be

$$\begin{aligned}
&\mathrm{Cov}[\tilde{v}_n, \tilde{v}_m] \\
&= M^2 \sum_{z\in\mathbb{Z}} \sum_{q\in\mathbb{Z}} c_{zN+n}(\varphi)\overline{c_{z-(\frac{m-n-kM}{N}+q\frac{L}{N})N+m}(\varphi)} \cdot \frac{\sigma^2}{M} \\
&= M\sigma^2 \sum_{q\in\mathbb{Z}} \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi)\overline{c_{zN+n+kM+qL}(\varphi)}.
\end{aligned} \tag{S38}$$

□

Appendix D
Appendix: maximum of correlation

Theorem 4. *Let $M, N \in \mathbb{N}_+^*$ and $n \in \mathbb{Z}$. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be two M-periodic functions. Denote the Fourier series of f and g by $\{c_z(f)\}_{z\in\mathbb{Z}}$ and $\{c_z(g)\}_{z\in\mathbb{Z}}$, respectively. Denote $\omega_M = e^{\frac{i2\pi}{M}}$. If the Fourier series of g is a subsequence of the Fourier series of f, i.e. $c_z(g) = c_{zN+n}(f)$ for any $z \in \mathbb{Z}$, then for any $x \in \mathbb{R}$, $g(x)$ can be expressed as:*

$$g(x) = \frac{1}{N}\omega_M^{-n\frac{x}{N}} \sum_{p=0}^{N-1} f(\frac{x}{N} + p\frac{M}{N})\omega_N^{-np} \tag{S39}$$

Proof. By definition, the Fourier series of f is

$$c_z(g) = \frac{1}{M}\int_0^M g(x)\,\omega_M^{-zx} dx \tag{S40}$$

And the expansion of $c_{zN+n}(f)$ is

$$\begin{aligned}
&c_{zN+n}(f) \\
&= \frac{1}{M}\int_0^M f(x)\,\omega_M^{-(zN+n)x} dx \\
&(\text{set } t = Nx) \\
&= \frac{1}{MN}\int_0^{MN} f(\frac{t}{N})\,\omega_M^{-(zN+n)\frac{t}{N}} dt \\
&= \frac{1}{MN}\sum_{p=0}^{N-1}\int_{pM}^{(p+1)M} f(\frac{t}{N})\,\omega_M^{-(zN+n)\frac{t}{N}} dt \\
&(\text{set } t' = t - pM \text{ for each term of } p) \\
&= \frac{1}{MN}\sum_{p=0}^{N-1}\int_0^M f(\frac{t'}{N} + p\frac{M}{N})\omega_M^{-n\frac{t'}{N}}\omega_N^{-np}\omega_M^{-zt'} dt' \\
&= \frac{1}{M}\int_0^M \underbrace{\frac{1}{N}\sum_{p=0}^{N-1} f(\frac{t'}{N} + p\frac{M}{N})\omega_M^{-n\frac{t'}{N}}\omega_N^{-np}}_{g(t')}\,\omega_M^{-zt'} dt'
\end{aligned} \tag{S41}$$

By identification from Eq. S41 and Eq. S40 we obtain

$$g(x) = \frac{1}{N}\omega_M^{-n\frac{x}{N}} \sum_{p=0}^{N-1} f(\frac{x}{N} + p\frac{M}{N})\omega_N^{-np} \tag{S42}$$

□

One can show that the constructed g in Eq. S39 is M-periodic.

Theorem 5. *Let $M, N \in \mathbb{N}_+^*$ and $k \in \mathbb{Z}$. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be an M-periodic real-valued function whose Fourier series is $\{c_z(\varphi)\}_{z\in\mathbb{Z}}$. Then the function $r(n) = \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi) \cdot \overline{c_{zN+n+kM}(\varphi)}$ for $n \in \mathbb{Z}$ can be expressed in the time domain by:*

$$r(n) = \frac{1}{MN}\sum_{\delta=0}^{N-1} r_\delta(n) \tag{S43}$$

where $r_\delta(n) = cos(\frac{2\pi\delta}{N}(\frac{kM}{2} + n))\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt$. Moreover, $n^* = -\frac{kM}{2}$ is a critical point of $r(n)$.

Proof. We start with calculating $r(n) = \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi)\cdot\overline{c_{zN+n+kM}(\varphi)}$. Let $A:\mathbb{R}\mapsto\mathbb{R}$ be an M-periodic function whose Fourier series $\{c_z(A)\}_{z\in\mathbb{Z}}$ satisfies $c_z(A) = c_{zN+n}(\varphi)$. According to Theorem 4,

$$A(x) = \frac{1}{N}\sum_{p=0}^{N-1}\varphi(\frac{x}{N}+p\frac{M}{N})\omega_M^{-n\frac{x}{N}}\omega_N^{-np} \tag{S44}$$

Likewise, let $B:\mathbb{R}\mapsto\mathbb{R}$ be an M-periodic function whose Fourier series $\{c_z(B)\}_{z\in\mathbb{Z}}$ satisfies $c_z(B) = c_{zN+n+kM}(\varphi)$, then

$$B(x) = \frac{1}{N}\sum_{p=0}^{N-1}\varphi(\frac{x}{N}+p\frac{M}{N})\omega_M^{-(n+kM)\frac{x}{N}}\omega_N^{-(n+kM)p} \tag{S45}$$

According to the Parseval's theorem, $\sum_{z\in\mathbb{Z}} c_z(A)\overline{c_z(B)} = \frac{1}{M}\int_0^M A(x)\overline{B(x)}dx$, it suffices to compute the integral on the right-hand side to compute $r(n)$.

$$\begin{aligned} r(n) &= \frac{1}{M}\int_0^M A(x)\overline{B(x)}dx \\ &= \frac{1}{MN^2}\int_0^M\sum_{p=0}^{N-1}\sum_{q=0}^{N-1}\varphi(\frac{x}{N}+p\frac{M}{N})\overline{\varphi(\frac{x}{N}+q\frac{M}{N})}\dots \\ &\qquad\cdot\omega_N^{kx}\omega_N^{n(q-p)+kMq}dx \\ &= \frac{1}{MN^2}\sum_{p=0}^{N-1}\sum_{q=0}^{N-1}\int_0^M\varphi(\frac{x}{N}+p\frac{M}{N})\varphi(\frac{x}{N}+q\frac{M}{N})\dots \\ &\qquad\cdot\omega_N^{k(x+qM)+n(q-p)}dx \\ &= \frac{1}{MN^2}\sum_{p=0}^{N-1}\sum_{q=0}^{N-1}\beta_{p,q}(n) \end{aligned} \tag{S46}$$

where

$$\beta_{p,q}(n) = \int_0^M\varphi(\frac{x}{N}+p\frac{M}{N})\varphi(\frac{x}{N}+q\frac{M}{N})\omega_N^{k(x+qM)+n(q-p)}dx \tag{S47}$$

Note that $\beta_{p,q}(n) = \beta_{p,q+N}(n)$, Eq. S46 can be rewritten as

$$\begin{aligned} r(n) &= \frac{1}{MN^2}\sum_{p=0}^{N-1}\sum_{q=p}^{p+N-1}\beta_{p,q}(n) \\ &= \frac{1}{MN^2}\sum_{p=0}^{N-1}\sum_{\delta=0}^{N-1}\beta_{p,p+\delta}(n) \\ &= \frac{1}{MN^2}\sum_{\delta=0}^{N-1}\sum_{p=0}^{N-1}\beta_{p,p+\delta}(n) \end{aligned} \tag{S48}$$

with $\delta = q-p$. Similarly, Eq. S46 can be rewritten as

$$r(n) = \frac{1}{MN^2}\sum_{\delta=0}^{N-1}\sum_{p=0}^{N-1}\beta_{p,p-\delta}(n) \tag{S49}$$

Averaging Eq. S48 and Eq. S49 amounts to

$$r(n) = \frac{1}{2MN^2}\sum_{\delta=0}^{N-1}\sum_{p=0}^{N-1}(\beta_{p,p-\delta}(n)+\beta_{p,p+\delta}(n)) \tag{S50}$$

We now focus on calculating the sum for each fixed δ. From Eq. S47 we have

$$\begin{aligned} &\sum_{p=0}^{N-1}\beta_{p,p+\delta}(n) \\ &= \sum_{p=0}^{N-1}\int_0^M\varphi(\frac{x}{N}+p\frac{M}{N})\varphi(\frac{x}{N}+(p+\delta)\frac{M}{N})\dots \\ &\qquad\cdot\omega_N^{k(x+(p+\delta)M)+n\delta}dx \\ &(\text{set } t = \frac{x}{N}+(p+\frac{\delta}{2})\frac{M}{N}) \\ &= N\sum_{p=0}^{N-1}\int_{(p+\frac{\delta}{2})\frac{M}{N}}^{(p+\frac{\delta}{2}+1)\frac{M}{N}}\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\dots \\ &\qquad\cdot\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \\ &= N\int_{\frac{\delta}{2}\frac{M}{N}}^{(N+\frac{\delta}{2})\frac{M}{N}}\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \\ &= N\int_{\frac{\delta}{2}\frac{M}{N}}^{M}\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \\ &+ N\int_M^{M+\frac{\delta}{2}\frac{M}{N}}\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \end{aligned} \tag{S51}$$

As $t\mapsto\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}$ is M-periodic,

$$\begin{aligned} &\sum_{p=0}^{N-1}\beta_{p,p+\delta}(n) \\ &= N\int_{\frac{\delta}{2}\frac{M}{N}}^{M}\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \\ &+ N\int_0^{\frac{\delta}{2}\frac{M}{N}}\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \\ &= N\int_0^M\varphi(t-\frac{\delta}{2}\frac{M}{N})\varphi(t+\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt+\frac{\delta}{2}M)+n\delta}dt \end{aligned} \tag{S52}$$

Likewise,

$$\begin{aligned} &\sum_{p=0}^{N-1}\beta_{p,p-\delta}(n) \\ &= N\int_0^M\varphi(t+\frac{\delta}{2}\frac{M}{N})\varphi(t-\frac{\delta}{2}\frac{M}{N})\omega_N^{k(Nt-\frac{\delta}{2}M)-n\delta}dt \end{aligned} \tag{S53}$$

Combining Eq. S52 with Eq. S53 amounts to

$$\begin{aligned} &\sum_{p=0}^{N-1}\beta_{p,p+\delta}(n)+\beta_{p,p-\delta}(n) \\ &= N\int_0^M\varphi(t+\frac{\delta}{2}\frac{M}{N})\varphi(t-\frac{\delta}{2}\frac{M}{N})\omega_N^{kNt}\dots \\ &\qquad\cdot(\omega_N^{\frac{\delta}{2}kM+n\delta}+\omega_N^{-\frac{\delta}{2}kM-n\delta})dt \\ &= 2N\cdot cos(\frac{2\pi\delta}{N}(\frac{kM}{2}+n))\dots \\ &\qquad\cdot\int_0^M\varphi(t+\frac{\delta}{2}\frac{M}{N})\varphi(t-\frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt \end{aligned} \tag{S54}$$

Combining Eq. S50 with Eq. S54, we finally obtain

$$r(n) = \frac{1}{MN} \sum_{\delta=0}^{N-1} r_\delta(n) \tag{S55}$$

where $r_\delta(n) := cos(\frac{2\pi\delta}{N}(\frac{kM}{2} + n)) \int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt$.

We remark that $r_{\delta=0}(n) = \int_0^M e^{i2\pi kt}\varphi(t)^2 dt$ is a constant independent of n, and $n^* = -\frac{kM}{2}$ is a critical point of $r_\delta(n)$ for $\delta \neq 0$. Thus $n^* = -\frac{kM}{2}$ is a critical point of $r(n)$. ☐

Corollary 1. *For a common interpolating filter φ which is even, $\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt \in \mathbb{R}$. If we admit $\int_0^M \varphi(t+\frac{\delta}{2}\frac{M}{N})\varphi(t-\frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt \geq 0$ for any $\delta \in [\![0, N-1]\!]$, then $r(n)$ is maximized at $n^* = -\frac{kM}{2}$. The maximum of $r(n)$ is*

$$\max_n r(n) = \frac{1}{MN} \sum_{\delta=0}^{N-1} \int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt. \tag{S56}$$

Corollary 2. *If φ is the nearest-neighbor filter and if $M \geq N$, then $r(n) = 0$.*

Proof. Indeed, the nearest-neighbor filter is expressed by $\varphi(t) = \mathbb{1}_{[-1/2,1/2]}(t)$. When $\delta = 0$,

$$\int_0^M \varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt = \int_{-1/2}^{1/2} 1 \cdot e^{i2\pi kt}dt = 0. \tag{S57}$$

When $\delta \in [\![1, N-1]\!]$, $\int_0^M \varphi(t+\frac{\delta}{2}\frac{M}{N})\varphi(t-\frac{\delta}{2}\frac{M}{N})e^{i2\pi kt}dt = 0$ because $\delta\frac{M}{N} \geq 1$ and $\varphi(t + \frac{\delta}{2}\frac{M}{N})\varphi(t - \frac{\delta}{2}\frac{M}{N}) = 0$ for any t.

Therefore, for any $\delta \in [\![0, N-1]\!]$ we have $r_\delta(n) = 0$, resulting in $r(n) = 0$. ☐

Corollary 2 describes the fact that any two frequency components in the spectrum of downsampled ($M \geq N$) *white noise* are uncorrelated when the nearest-neighbor filter is used for downsampling. Indeed, a white noise after downsampling using the nearest-neighbor filter is simply a subsequence of the original white noise, which is still a white noise. Hence, the frequency components in the output spectrum are uncorrelated.

However, in the case of downsampling a signal whose power spectrum density is not homogeneous using the nearest-neighbor filter, there exist pairs of frequency components again in the spectrum of the downsampled signal that are correlated. Fig. 9 showing very high detecting accuracies on images downsampled with the nearest-neighbor filter also confirms this fact.

Appendix E
Theorem: Spectrum Folding of Non-Aligned Resampling

The previous theorems describe the spectrum folding of resampling a discrete signal from the entire interpolated function. In practice, however, it is common to process the resampling on an interval in a length that is different from the period of the interpolated function. For instance, sometimes the corner points are required to be aligned[4] when resampling a signal. In such a case, the interval of discretizing the interpolated function is not exactly in alignment with the support of the interpolated function. This would slightly alter the phenomenon of spectrum folding and change the distance between correlated frequency components in the spectrum.

Consider again a 1D signal $\{u_k\} \in \mathbb{R}^M$ of M samples and its resampled version $\{v_k\} \in \mathbb{R}^N$, with $M, N \in \mathbb{N}^*$ being any two positive integers. The non-aligned resampling is modeled in two steps: first the samples $\{u_k\}$ are interpolated to a continuous function f using an M-periodic interpolating filter φ in the same way as the aligned-resampling, then f is resampled to the discrete signal $\{v_k\}$ of N samples on a support in length M' that is slightly different from M.

Again, the continuous function $f : \mathbb{R} \to \mathbb{R}$ as the output of the interpolation is $f(x) = \sum_{k=0}^{M-1} u_k\, \varphi(x - k)$.

For the non-aligned resampling, let the size of the resampling interval be $M' = M + \Delta$ with $\Delta \in [-1, 1]$, and f is discretized evenly on $[0, M']$ to sequence $\mathbf{v} = \{v_k\} \in \mathbb{R}^N$ by taking $v_k = f(k\frac{M'}{N})$. Then in the spectrum of the resampled signal, two frequency components separated by $M-1$, M or $M+1$ components are correlated.

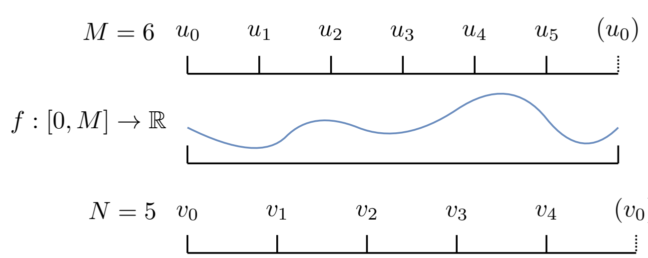

Fig. S2: Non-aligned resampling: the intervals for the original sequence and the resampled sequence are slightly different.

Theorem 6. *Suppose the intepolating filter φ is M-periodic. Denote the DFTs of $\{u_k\}$ and $\{v_k\}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier coefficients of φ and f by $\{c_p(\varphi)\}_p$ and $\{c_p(f)\}_p$, respectively.*

Define a 2D series $T_{p,q}$ by

$$T_{p,q} = -ie^{\pi i M'(\frac{q}{M} - \frac{p}{M'})} sinc\left(\pi M'\left(\frac{q}{M} - \frac{p}{M'}\right)\right).$$

Then:

- *if $\Delta = 0$, we have*

$$\tilde{v}_n = M \sum_{z\in\mathbb{Z}} c_{zN+n}(\varphi)\tilde{u}_{zN+n} \tag{S58}$$

[4] E.g. in the libraries PyTorch and Tensorflow the resizing function provides an option of aligning the corner points of the input and output signals.

- *if* $\Delta \in (0,1]$, *setting* $\alpha_p := c_p(\varphi)T_{p+1,p}$ *and* $\beta_p := c_p(\varphi)T_{p,p}$, *we have*
$$\tilde{v}_n \approx M \sum_{z\in\mathbb{Z}} \alpha_{zN+n-1}\, \tilde{u}_{zN+n-1} + \beta_{zN+n}\, \tilde{u}_{zN+n} \tag{S59}$$
- *if* $\Delta \in [-1,0)$, *setting* $\alpha'_p := c_p(\varphi)T_{p,p}$ *and* $\beta'_p := c_p(\varphi)T_{p-1,p}$, *we have*
$$\tilde{v}_n \approx M \sum_{z\in\mathbb{Z}} \alpha'_{zN+n}\, \tilde{u}_{zN+n} + \beta'_{zN+n+1}\, \tilde{u}_{zN+n+1} \tag{S60}$$

Proof. Again, denote the DFTs of $\mathbf{u}$ and $\mathbf{v}$ by $\{\tilde{u}_m\} \in \mathbb{C}^M$ and $\{\tilde{v}_n\} \in \mathbb{C}^N$, respectively. Denote the Fourier series of φ and f by $\{c_p(\varphi)\}_{p\in\mathbb{Z}}$ and $\{c_p(f)\}_{p\in\mathbb{Z}}$, respectively.

In Sec. A, we have shown $c_p(f) = M \cdot c_p(\varphi) \cdot \tilde{u}_p$ for the interpolation process.

The next step is to link $\{\tilde{v}_m\} \in \mathbb{C}^N$ to $\{c_p(f)\}_{p\in\mathbb{Z}}$ for the discretization process. For this we need to construct a M'-periodic bridge function $g : \mathbb{R} \to \mathbb{R}$ which coincides with f on $[0, M']$, *i.e.* $g\big|_{[0,M']} = f\big|_{[0,M']}$, then resample $\{v_k\}$ from g by taking $v_k = g(k\frac{M'}{N})$. The Fourier series $\{c_p(g)\}_{p\in\mathbb{Z}}$ of g satisfies

$$\begin{aligned}
c_p(g) &= \frac{1}{M'} \int_0^{M'} g(x)\, \omega_{M'}^{-px}\, dx \\
&= \frac{1}{M'} \int_0^{M'} f(x)\, \omega_{M'}^{-px}\, dx \\
&= \frac{1}{M'} \int_0^{M'} \left(\sum_{q\in\mathbb{Z}} c_q(f)\, \omega_M^{qx} \right) \omega_{M'}^{-px}\, dx \\
&= \frac{1}{M'} \int_0^{M'} \sum_{q\in\mathbb{Z}} c_q(f)\, \omega_M^{qx}\, \omega_{M'}^{-px}\, dx \\
&= \frac{1}{M'} \sum_{q\in\mathbb{Z}} \int_0^{M'} c_q(f)\, \omega_M^{qx}\, \omega_{M'}^{-px}\, dx \qquad (*) \\
&= \frac{1}{M'} \sum_{q\in\mathbb{Z}} c_q(f) \int_0^{M'} e^{2\pi i x\left(\frac{q}{M} - \frac{p}{M'}\right)}\, dx \\
&\triangleq \sum_{q\in\mathbb{Z}} c_q(f)\, T_{p,q}
\end{aligned} \tag{S61}$$

with

$$\begin{aligned}
T_{p,q} &= \frac{1}{M'} \int_0^{M'} e^{2\pi i x\left(\frac{q}{M} - \frac{p}{M'}\right)}\, dx \\
&= -ie^{\pi i M'\left(\frac{q}{M} - \frac{p}{M'}\right)} \mathrm{sinc}\left(\pi M' \left(\frac{q}{M} - \frac{p}{M'}\right)\right)
\end{aligned} \tag{S62}$$

Using $M' = M + \Delta$, we have

$$T_{p,q} = -ie^{i\pi\left(p - \left(\frac{\Delta}{M}+1\right)q\right)} \mathrm{sinc}\left(\pi \left(p - \left(\frac{\Delta}{M} + 1\right) q\right)\right) \tag{S63}$$

$$|T_{p,q}| = \mathrm{sinc}\left(\pi \left(p - \left(\frac{\Delta}{M} + 1\right) q\right)\right) \tag{S64}$$

The equality at line (*) applies the Fubini's theorem based on the fact that $\int_0^{M'} \sum_{q\in\mathbb{Z}} |c_q(f)\, \omega_M^{qx}\, \omega_{M'}^{-px}|\, dx = M' \sum_{q\in\mathbb{Z}} |c_q(f)| < +\infty$.

Eq. S61 describes each Fourier coefficient of g as a weighted sum of Fourier coefficients of f. The weights $T_{p,q}$ having a sinc term are negligible when $\pi M' |\frac{q}{M} - \frac{p}{M'}|$ is large. For a fixed p, we are only interested in $T_{p,q}$ whose magnitude is large. Actually, $q \mapsto |T_{p,q}|$ is maximized at $q_{max} = p\left(1 - \frac{\Delta}{M+\Delta}\right)$. We additionally limit $p \in (-M, M)$ since the Fourier coefficient $|c_p(g)|$ decreases rapidly as $|p|$ increases.

If $\Delta = 0$, $T_{p,q} = 1$ when $q = p$ and $T_{p,q} = 0$ when $q \neq p$, thus $c_p(g) = c_p(f)$. The case reduces to the aligned resampling.

If $\Delta \in (0, 1]$, $q_{max} \in [p-1, p)$. As q must be an integer, the largest two coefficients $T_{p,q}$ correspond to $q = p-1$ and $q = p$. Eq. S61 can be approximated by

$$c_p(g) \approx c_{p-1}(f)\, T_{p,p-1} + c_p(f)\, T_{p,p} \tag{S65}$$

Resampling the continuous g at sampling rate $\frac{M'}{N}$ to obtain the discrete sequence $\{v_k\} \in \mathbb{R}^N$ such that $v_k = g(k\frac{M'}{N})$ is equivalent to summing up the non-overlapped intervals in length N of the Fourier transform of g [44]:

$$\begin{aligned}
\tilde{v}_n &= \sum_{z\in\mathbb{Z}} c_{zN+n}(g) \\
&\approx \sum_{z\in\mathbb{Z}} c_{zN+n-1}(f)\, T_{zN+n,zN+n-1} + \dots \\
&\qquad\qquad c_{zN+n}(f)\, T_{zN+n,zN+n} \\
&= M \sum_{z\in\mathbb{Z}} c_{zN+n-1}(\varphi)\, T_{zN+n,zN+n-1}\, \tilde{u}_{zN+n-1} \\
&\qquad + c_{zN+n}(\varphi)\, T_{zN+n,zN+n}\, \tilde{u}_{(zN+n)}
\end{aligned} \tag{S66}$$

Setting $\alpha_p := c_p(\varphi)\, T_{p+1,p}$ and $\beta_p := c_p(\varphi)\, T_{p,p}$, we have

$$\tilde{v}_n = M \sum_{z\in\mathbb{Z}} \alpha_{zN+n-1}\, \tilde{u}_{(zN+n-1)} + \beta_{zN+n}\, \tilde{u}_{zN+n} \tag{S67}$$

Similarly, if $\Delta \in [-1, 0)$, $q_{max} \in (p, p+1]$, Eq. S61 can be approximated by

$$c_p(g) \approx c_p(f)\, T_{p,p} + c_{p+1}(f)\, T_{p,p+1} \tag{S68}$$

Setting $\alpha'_p := c_p(\varphi)\, T_{p,p}$ and $\beta'_p := c_p(\varphi)\, T_{p-1,p}$, we have

$$\tilde{v}_n = M \sum_{z\in\mathbb{Z}} \alpha'_{zN+n} \tilde{u}_{zN+n} + \beta'_{zN+n+1} \tilde{u}_{zN+n+1} \tag{S69}$$

□

Theorem 6 expresses the relationship between the input DFT, the Fourier coefficients of the interpolating filter and the output DFT in the case of non-aligned resampling. In particular, let $k \in \mathbb{Z}$, one can remark that when $\Delta \neq 0$, $\tilde{v}_n$ is likely to be correlated with $\tilde{v}_{n+kM-1}$, $\tilde{v}_{n+kM}$ and $\tilde{v}_{n+M+1}$. Indeed, if $\Delta \in (0, 1]$, we have

$$\tilde{v}_n \approx M \sum_{z\in\mathbb{Z}} \alpha_{zN+n-1}\, \tilde{u}_{zN+n-1} + \beta_{zN+n}\, \tilde{u}_{zN+n} \tag{S70}$$

and

$$\begin{aligned}
\tilde{v}_{n+kM-1} &\approx M \sum_{z\in\mathbb{Z}} \alpha_{zN+n+kM-2}\, \tilde{u}_{zN+n-2} + \dots \\
&\qquad \beta_{zN+n+kM-1}\, \tilde{u}_{zN+n-1} \\
\tilde{v}_{n+kM} &\approx M \sum_{z\in\mathbb{Z}} \alpha_{zN+n+kM-1}\, \tilde{u}_{zN+n-1} + \dots \\
&\qquad \beta_{zN+n+kM}\, \tilde{u}_{zN+n} \\
\tilde{v}_{n+kM+1} &\approx M \sum_{z\in\mathbb{Z}} \alpha_{zN+n+kM}\, \tilde{u}_{zN+n} + \dots \\
&\qquad \beta_{zN+n+kM+1}\, \tilde{u}_{zN+n+1}
\end{aligned} \tag{S71}$$

which all involve the same terms $\tilde{u}_{zN+n-1}$ and $\tilde{u}_{zN+n}$ in the weighted sum.

Likewise, if $\Delta \in [-1, 0)$, we have

$$\tilde{v}_n \approx M \sum_{z\in\mathbb{Z}} \alpha'_{zN+n}\, \tilde{u}_{zN+n} + \beta'_{zN+n+1}\, \tilde{u}_{zN+n+1} \tag{S72}$$

and

$$\begin{aligned}
\tilde{v}_{n+kM-1} &\approx M \sum_{z\in\mathbb{Z}} \alpha'_{zN+n+kM-1}\, \tilde{u}_{zN+n-1} + \dots \\
&\qquad \beta'_{zN+n+kM}\, \tilde{u}_{zN+n} \\
\tilde{v}_{n+kM} &\approx M \sum_{z\in\mathbb{Z}} \alpha'_{zN+n+kM}\, \tilde{u}_{zN+n} + \dots \\
&\qquad \beta'_{zN+n+kM+1}\, \tilde{u}_{zN+n+1} \\
\tilde{v}_{n+kM+1} &\approx M \sum_{z\in\mathbb{Z}} \alpha'_{zN+n+kM+1}\, \tilde{u}_{zN+n+1} + \dots \\
&\qquad \beta'_{zN+n+kM+2}\, \tilde{u}_{zN+n+2}
\end{aligned} \tag{S73}$$

which all involve the same terms $\tilde{u}_{zN+n}$ and $\tilde{u}_{zN+n+1}$ in the weighted sum.

Therefore, in this case of resampling a signal of M samples into N samples with slightly non-aligned resampling interval, two frequency components separated by $kM-1$, kM or $kM+1$ components in the spectrum of the resampled signal are likely to be correlated. Moreover, as $|k|$ increases, the correlation vanishes rapidly. This corresponds to the observation in Fig. 2 that the significant distances d related to anomalous spectral correlations belong to $[\![M-1 (\mathrm{mod} N), M+1 (\mathrm{mod}\ N)]\!] \cup [\![-M-1 (\mathrm{mod} N), -M+1 (\mathrm{mod} N)]\!]$.

Appendix F
Implementation and Computational Cost

The most time-consuming part of the *a contrario* algorithm in Sec. III-B is the computation of the correlations between non-overlapped patches and their shifted counterparts in the Fourier spectrum. More precisely, given an image in $H \times W$, its Fourier spectrum is divided into non-overlapped patches in $h \times w$, and the correlations between each patch and those at vertical distances $d \in [\![0, H-1]\!]$ are computed, resulting in a total of $M \times N \times H$ pair-wise correlations where $M = \lfloor \frac{H}{h} \rfloor$ and $N = \lfloor \frac{W}{w} \rfloor$. The naive computation of all the pair-wise correlations has time complexity $\mathcal{O}(MNH \times hw) = \mathcal{O}(H^2 W)$.

Recall that the Pearson Correlation Coefficient between two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ is computed in the form of

$$corr(\mathbf{x}, \mathbf{y}) := \frac{\sum x_i \bar{y}_i - \frac{1}{n} \sum x_i \overline{\sum y_i}}{\sqrt{\sum |x_i|^2 - \frac{|\sum x_i|^2}{n}} \sqrt{\sum |y_i|^2 - \frac{|\sum y_i|^2}{n}}} \tag{S74}$$

Two optimizations are implemented: 1) the sum $\sum x_i$ and the energy $\sum |x_i|^2$ can be computed efficiently using integral images [49], reducing the time complexity from $\mathcal{O}(H \times W \times h \times w)$ to $\mathcal{O}(H \times W)$; 2) the inner products $\sum x_i \bar{y}_i$ between each anchor patch $\mathbf{x}$ and its shifted counterparts $\mathbf{y}$ at distances $d \in [\![0, H-1]\!]$ can be computed efficiently using Fast Fourier Transform (FFT).

Integral Image Let $\mathbf{F} \in \mathbb{C}^{H \times W}$ be the Fourier spectrum of an $H \times W$ image. Let $\mathcal{I}(\mathbf{F})$ and $\mathcal{I}(|\mathbf{F}|^2)$ be the integral images of $\mathbf{F}$ and $|\mathbf{F}|^2$, respectively:

$$\begin{aligned}
\mathcal{I}(\mathbf{F})(u, v) &:= \sum_{i=0}^{u} \sum_{j=0}^{v} \mathbf{F}(i, j) \\
\mathcal{I}(|\mathbf{F}|^2)(u, v) &:= \sum_{i=0}^{u} \sum_{j=0}^{v} |\mathbf{F}(i, j)|^2
\end{aligned} \tag{S75}$$

with boundary values defined as $\mathcal{I}(\mathbf{F})(-1, v) = \mathcal{I}(\mathbf{F})(u, -1) = 0$ and $\mathcal{I}(|\mathbf{F}|^2)(-1, v) = \mathcal{I}(|\mathbf{F}|^2)(u, -1) = 0$ for all u and v. The integral images $\mathcal{I}(\mathbf{F})$ and $\mathcal{I}(|\mathbf{F}|^2)$ can be computed in one pass over $\mathbf{F}$ with time complexity $\mathcal{O}(HW)$.

Given a rectangular patch $\mathbf{P}$ in $\mathbf{F}$ defined by its top-left and bottom-right coordinates (u_1, v_1) and (u_2, v_2), its sum can be computed efficiently from the integral image $\mathcal{I}(\mathbf{F})$ with constant time complexity as follows:

$$\begin{aligned}
S(\mathbf{P}) &:= \sum_{u=u_1}^{u_2} \sum_{v=v_1}^{v_2} \mathbf{F}(u, v) \\
&= \mathcal{I}(\mathbf{F})(u_2, v_2) + \mathcal{I}(\mathbf{F})(u_1 - 1, v_1 - 1) \\
&\quad - \mathcal{I}(\mathbf{F})(u_2, v_1 - 1) - \mathcal{I}(\mathbf{F})(u_1 - 1, v_2)
\end{aligned} \tag{S76}$$

Similarly, the energy of the patch $\mathbf{P}$ can be computed using $\mathcal{I}(|\mathbf{F}|^2)$:

$$\begin{aligned}
E(\mathbf{P}) &:= \sum_{u=u_1}^{u_2} \sum_{v=v_1}^{v_2} |\mathbf{F}(u, v)|^2 \\
&= \mathcal{I}(|\mathbf{F}|^2)(u_2, v_2) + \mathcal{I}(|\mathbf{F}|^2)(u_1 - 1, v_1 - 1) \\
&\quad - \mathcal{I}(|\mathbf{F}|^2)(u_2, v_1 - 1) - \mathcal{I}(|\mathbf{F}|^2)(u_1 - 1, v_2)
\end{aligned} \tag{S77}$$

And the standard deviation of the patch $\mathbf{P}$ is computed as

$$\begin{aligned}\sigma(\mathbf{P}) &:= \sqrt{\sum_{u=u_1}^{u_2}\sum_{v=v_1}^{v_2}|\mathbf{F}(u,v)|^2 - \frac{|\sum_{u=u_1}^{u_2}\sum_{v=v_1}^{v_2}\mathbf{F}(u,v)|^2}{(u_2-u_1+1)(v_2-v_1+1)}} \\ &= \sqrt{E(\mathbf{P}) - \frac{|S(\mathbf{P})|^2}{(u_2-u_1+1)(v_2-v_1+1)}}\end{aligned} \tag{S78}$$

FFT-based Computation of Correlations Let $\{\mathbf{P}_{mn} \in \mathbb{C}^{h\times w}, (m,n) \in [\![0, M-1]\!]\times[\![0, N-1]\!]\}$ be the non-overlapped anchor patches of $\mathbf{F}$:

$$\mathbf{P}_{mn}(p,q) := \mathbf{F}(mh+p, nw+q),\ 0 \le p < h, 0 \le q < w \tag{S79}$$

The shifted counterpart of $\mathbf{P}_{mn}$ at distance d along the vertical axis is denoted by $\mathbf{P}_{mn}^{(d)}$:

$$\mathbf{P}_{mn}^{(d)}(p,q) := \mathbf{F}(mh+p+d, nw+q),\ 0 \le p < h, 0 \le q < w \tag{S80}$$

The Pearson Correlation Coefficient between $\mathbf{P}_{mn}$ and $\mathbf{P}_{mn}^{(d)}$ is computed as:

$$\begin{aligned}&corr(\mathbf{P}_{mn}, \mathbf{P}_{mn}^{(d)}) \\ &= \frac{\sum_{p=0}^{h-1}\sum_{q=0}^{w-1}\mathbf{P}_{mn}(p,q)\overline{\mathbf{P}_{mn}^{(d)}(p,q)} - \frac{1}{hw}S(\mathbf{P}_{mn})\overline{S(\mathbf{P}_{mn}^{(d)})}}{\sigma(\mathbf{P}_{mn})\sigma(\mathbf{P}_{mn}^{(d)})} \\ &= \frac{R_{mn}(d) - \frac{S(\mathbf{P}_{mn})\overline{S(\mathbf{P}_{mn}^{(d)})}}{hw}}{\sigma(\mathbf{P}_{mn})\sigma(\mathbf{P}_{mn}^{(d)})}\end{aligned} \tag{S81}$$

where $R_{mn}(d) := \sum_{p=0}^{h-1}\sum_{q=0}^{w-1}\mathbf{P}_{mn}(p,q)\overline{\mathbf{P}_{mn}^{(d)}(p,q)}$ is the inner product between the two patches. Computing $R_{mn}(d)$ for distances $d \in [\![0, H-1]\!]$ is equivalent to computing the cross-correlation between the anchor patch $\mathbf{P}_{mn}$ and the vertical band formed by the columns of $\mathbf{F}$ from nw to $nw+w-1$.

$$R_{mn}(d) = \sum_{q=0}^{w-1}\sum_{p=0}^{h-1}\mathbf{P}_{mn}(p,q)\overline{\mathbf{F}(mh+p+d, nw+q)} \tag{S82}$$

If we extend $\mathbf{P}_{mn}$ by zero-padding to size $H\times w$, denoted by $\widehat{\mathbf{P}}_{mn} \in \mathbb{C}^{H\times w}$:

$$\widehat{\mathbf{P}}_{mn}(u,q) := \begin{cases}\mathbf{P}_{mn}(u,q), & \text{if } 0 \le u < h \\ 0, & \text{otherwise}\end{cases} \tag{S83}$$

Then we have

$$R_{mn}(d) = \sum_{q=0}^{w-1}\sum_{p=0}^{H-1}\widehat{\mathbf{P}}_{mn}(p,q)\overline{\mathbf{F}(mh+p+d, nw+q)} \tag{S84}$$

The inner sum for a fixed q can be efficiently computed using the FFT, thus

$$\begin{aligned}&R_{mn}(d) \\ &= \sum_{q=0}^{w-1}\mathcal{F}^{-1}\left[\overline{\mathcal{F}\left\{\widehat{\mathbf{P}}_{mn}(\cdot,q)\right\}}\odot\mathcal{F}\{\mathbf{F}(\cdot,nw+q)\}\odot e^{\frac{2\pi i}{H}mhk}\right](d)\end{aligned} \tag{S85}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the FFT and IFFT, k is the frequency index, and $\odot$ is the element-wise multiplication. This is much faster because it suffices to pre-compute the FFT of each column of $\mathbf{F}$ only once and reuse them for all anchor patches.

Finally, the total time complexity of computing all the correlations between the anchor patches and their shifted counterparts is reduced to $\mathcal{O}(MNwH\log(H)) = \mathcal{O}(MHW\log(H))$. Compared to the naive computation with time complexity $\mathcal{O}(H^2W)$, the FFT-based implementation is accelerated by a factor of $\mathcal{O}(\frac{H}{M\log(H)}) = \mathcal{O}(\frac{h}{\log(H)})$ with h the height of the patches. This is beneficial when h is large, i.e. when using large patches.

The complete pseudocode of the optimized implementation of the *a contrario* detection method along the vertical axis is detailed in Alg. 1.

Algorithm 1: NFA computation for detecting anomalous correlations along the vertical axis

Input : $\mathbf{F} \in \mathbb{C}^{H\times W}$, Fourier spectrum
Input : Patch size (h,w) and local maximum radius r
Output : NFAs for vertical shifts $d \in [\![r+1, H-r-1]\!]$

```
M ← ⌊H/h⌋, N ← ⌊W/w⌋
Compute integral images 𝓘(F) and 𝓘(|F|²) (Eq. S75)
Compute column-wise DFTs 𝓕{F(·,v)} for all v
for (m,n) ∈ ⟦0, M−1⟧ × ⟦0, N−1⟧ do
    Compute 𝓕{P̂_mn(·,q)} for all q ∈ ⟦0, w−1⟧ (Eq. S83)
    Compute R_mn(d) for all d ∈ ⟦0, H−1⟧ via FFT-based inner products (Eq. S85)
    for d ∈ ⟦1, H−1⟧ do
        Compute S(P_mn), S(P_mn^(d)), σ(P_mn), σ(P_mn^(d)) using integral images (Eqs. S76, S77, S78)
        ρ(m,n,d) ← corr(P_mn, P_mn^(d)) (Eqs. S81, S83, S85)
Initialize NFA(d) ← +∞ for all d ∈ ⟦r+1, H−r−1⟧
for d ∈ ⟦r+1, H−r−1⟧ do
    k ← #{(m,n) : ρ(m,n,d) = max_{d'∈⟦d−r,d+r⟧} ρ(m,n,d')}
    NFA(d) ← (H − 2r − 1) B(k, MN, 1/(2r+1)), B is the tail of binomial law
return NFA
```

In our implementation, the patch size (h,w) is set to $\frac{1}{10}$ of the image dimensions along each axis. The local maximum radius is fixed to $r = 20$. For the residual extraction stage, the rank transform is applied using a 7×7 local neighborhood, and the TV denoising step follows the method described in [47], with the regularization parameter set to $\lambda = 1$.

The computational cost was evaluated on an Apple Mac M1 Pro equipped with a 3.2 GHz CPU and a memory bandwidth of 200 GB/s. The method was implemented in Python 3.10.19. The computation times of the algorithm for different image sizes and residual extraction filters are reported in Fig. S3. The results show that the most time-consuming component is the residual extraction step with TV denoising. In contrast, the algorithm without residual extraction or using the rank transform requires less than 1.5 seconds even for large images of size 2000×2000.

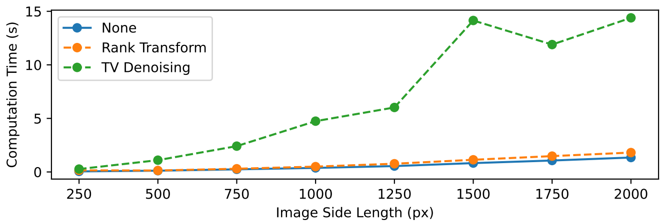

Fig. S3: Computation time of the proposed method for different image sizes and residual extraction filters.

Appendix G
ROC curves of different unsupervised detection methods

The receiver operating curves (ROC) and the areas under the curve (AUC) of the comparison experiments on images without anti-aliasing in Sec. IV-A are shown in Fig. S4. Each curve counts the original images as true samples and all resampled images at all resampling factors between 0.6 and 1.6 as false samples. We can see that our method outperforms the others by a large margin, having high recalls of about 100%, 90% and 80%, respectively for the three compression configurations at very low false positive rates.

Appendix H
Additional qualitative results

We present additional qualitative results of the proposed method with different residual extraction filters for resampling detection on resampled images saved in uncompressed PNG format. All the analyzed images were resampled from 1000×1000 to 400×400 and saved in PNG format. Resampling detection was performed along the horizontal axis.

Fig. S5 shows an ideal case of Gaussian noise. The *a contrario* algorithm is able to detect the spectral correlations of the underlying resampling even without residual extraction, and also with any other residual extraction filters.

Fig. S6 shows the performance on a real image containing fine-grained textures with weak contrasts. As for the case of resampled Gaussian noise, the proposed method performs correctly without residual extraction or using any other residual extraction filters.

In contrast, Fig. S7 presents a challenging natural image characterized by highly contrasted structures. The inherent vertical components—such as tree trunks, window frames, and building edges—are replicated horizontally, inducing strong spectral correlations that mask the subtle traces of resampling. In this case, only the rank transform filter effectively extracts the residuals necessary to reveal the hidden resampling artifacts.

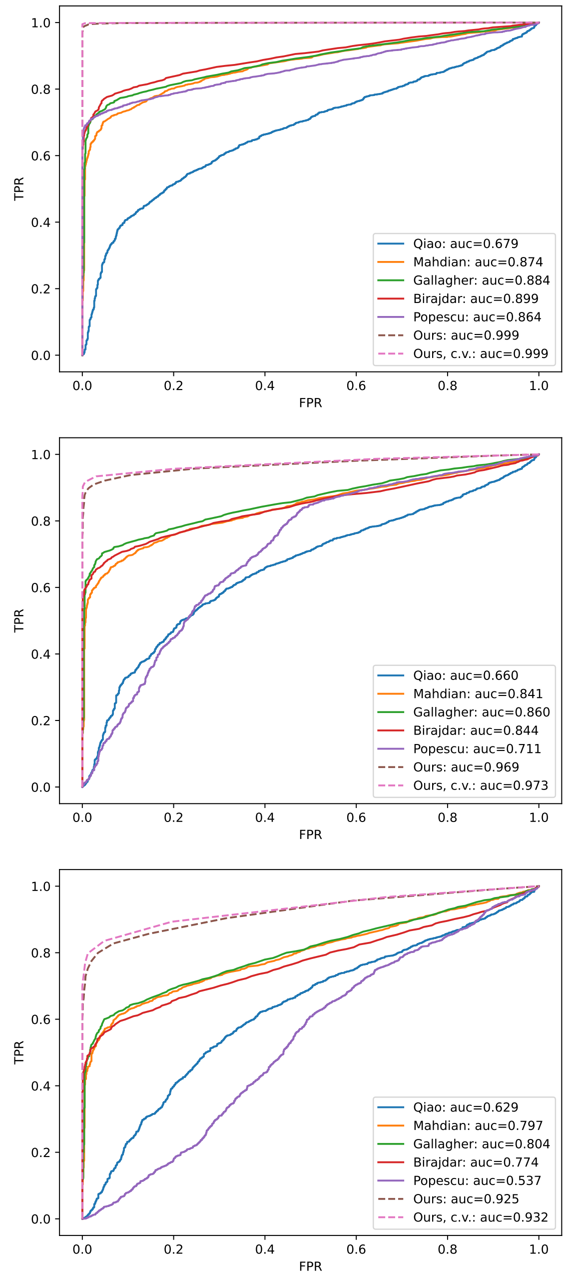

Fig. S4: ROC curves and AUCs of different resampling detection methods on images resampled at resampling rates from 0.6 to 1.6. Left: the images are saved in PNG; middle: the images are compressed by JPEG at quality factor 95; right: the images are compressed by JPEG at quality factor 90. Our method is highlighted in dashed curves.

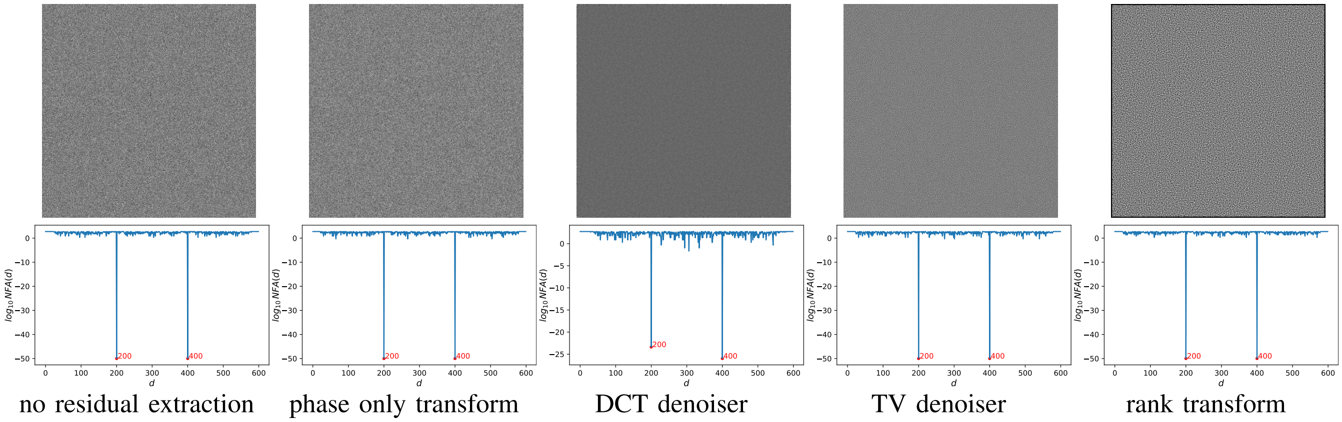

Fig. S5: Detection results with different residual extraction filters on a Gaussian noise image resampled from 1000×1000 to 600×600. Top: images after residual extraction; bottom: the NFAs of different distances d. The expected positive correlation distances d at 200 and 400 are detected using all residual extraction filters.

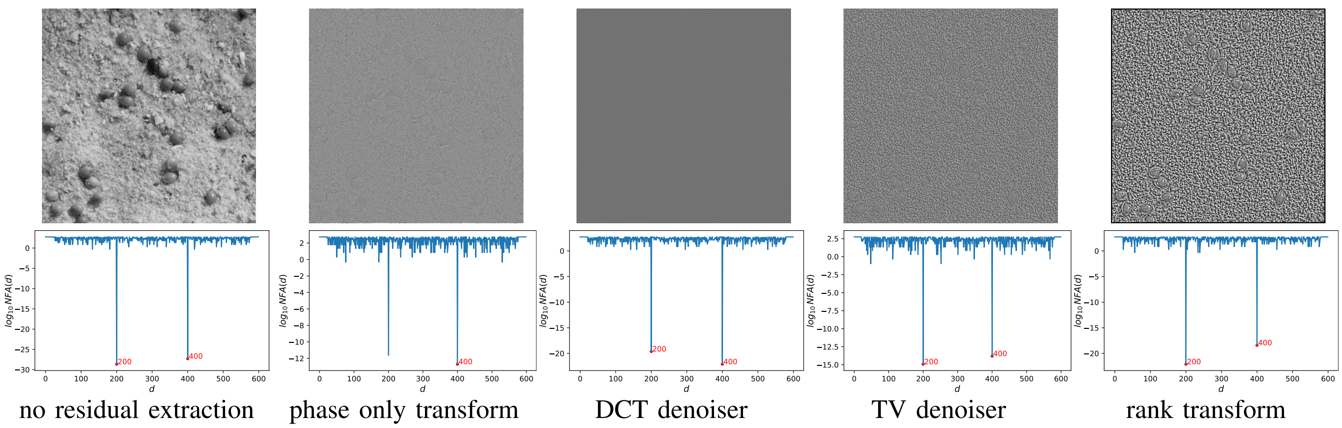

Fig. S6: Detection results with different residual extraction filters on an image with fine-grained textures resampled from 1000×1000 to 600×600. Top: images after residual extraction; bottom: the NFAs of different distances d. The expected positive correlation distances d at 200 and 400 are detected using all residual extraction filters.

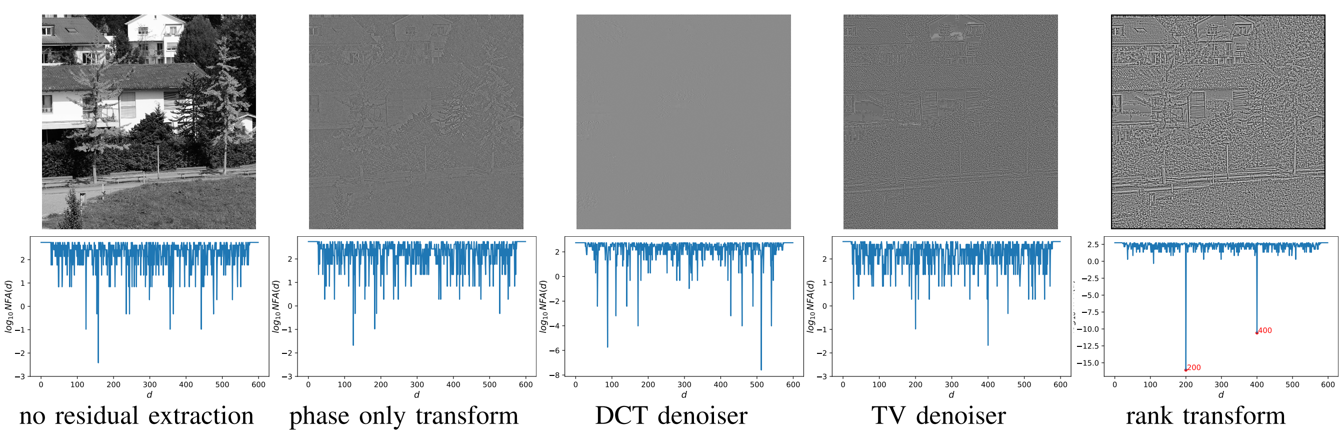

Fig. S7: Detection results with different residual extraction filters on an image with contrasted and fine-grained textures resampled from 1000×1000 to 600×600. Top: images after residual extraction; bottom: the NFAs of different distances d. The expected positive correlation distances d at 200 and 400 are detected with the rank transform only.